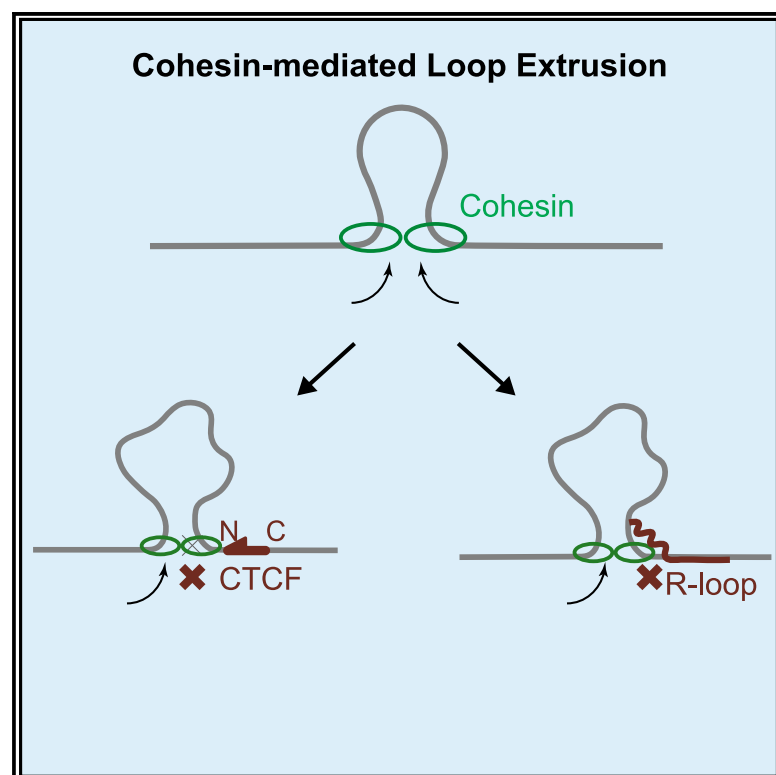


CTCF and R-loops are boundaries of cohesin-mediated DNA looping

Graphical abstract



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In brief

CTCF and cohesin organize the 3D eukaryotic genome. Zhang et al. report direct single-molecule and structure evidence that CTCF is a polar cohesin barrier. R-loops are also cohesin barriers and likely play a role in organizing the 3D genome.

Highlights

- CTCF is a polar barrier to cohesin-mediated DNA looping
- A structure of the cohesin-NIPBL-CTCF-DNA complex reveals CTCF-cohesin interactions
- R-loops slow cohesin translocation *in vitro* and are enriched with cohesin *in vivo*



Article

CTCF and R-loops are boundaries of cohesin-mediated DNA looping

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SUMMARY

Cohesin and CCCTC-binding factor (CTCF) are key regulatory proteins of three-dimensional (3D) genome organization. Cohesin extrudes DNA loops that are anchored by CTCF in a polar orientation. Here, we present direct evidence that CTCF binding polarity controls cohesin-mediated DNA looping. Using single-molecule imaging, we demonstrate that a critical N-terminal motif of CTCF blocks cohesin translocation and DNA looping. The cryo-EM structure of the cohesin-CTCF complex reveals that this CTCF motif ahead of zinc fingers can only reach its binding site on the STAG1 cohesin subunit when the N terminus of CTCF faces cohesin. Remarkably, a C-terminally oriented CTCF accelerates DNA compaction by cohesin. DNA-bound Cas9 and Cas12a ribonucleoproteins are also polar cohesin barriers, indicating that stalling may be intrinsic to cohesin itself. Finally, we show that RNA-DNA hybrids (R-loops) block cohesin-mediated DNA compaction *in vitro* and are enriched with cohesin subunits *in vivo*, likely forming TAD boundaries.

INTRODUCTION

Higher eukaryotes fold their genomes into topologically associating domains (TADs).^{1–5} DNA sequences within a TAD interact frequently with each other but are insulated from adjacent TADs. The cohesin complex, which is constituted of SMC1, SMC3, RAD21, and either STAG1 or STAG2, and CCCTC-binding factor (CTCF) are both enriched at TAD boundaries.^{1–3,6–12} Depleting CTCF or cohesin disrupts chromosomal looping and insulation between most TADs.^{13–17} CTCF defines TAD boundaries by blocking the loop extrusion activity of cohesin via an incompletely understood mechanism.^{18,19} TADs are also established via CTCF-independent mechanisms, including transcription and replication activities that restrict cohesin loop extrusion.^{20–23} The mechanisms underlying cohesin regulation at these roadblocks remain unclear. Here, we explore the mechanisms of CTCF-dependent and -independent cohesin arrest during loop extrusion to shape the three-dimensional (3D) genome.

CTCF arrests cohesin in an orientation-specific manner in most higher eukaryotes.^{10,11,24–26} TAD boundaries are marked by CTCF-binding sites (CBSs) in a convergent arrangement. De-

leting genomic CBSs abrogates TAD boundaries^{11,24,25} and induces aberrant gene activation.^{27–29} An interaction between an N-terminal CTCF peptide with a Tyr-Asp-Phe (YDF) motif and cohesin STAG1/2 subunit is essential for polar cohesin arrest and for maintaining TAD boundaries *in vivo*.¹⁹ However, the mechanisms regulating polar cohesin arrest remain poorly explored due to the difficulty of reconstituting these biochemical activities for structure-function studies.

Topological boundaries are also established via non-CTCF mechanisms such as replication and transcription. Notably, CTCF demarcates <10% of TADs in fruit flies in an orientation-independent manner.^{3,6,30,31} Fly TADs are depleted in active chromatin marks and separated by regions of active chromatin.³⁰ The minichromosome maintenance (MCM) complex can impede the formation of CTCF-anchored cohesin loops and TADs in a cell-cycle-specific manner.²³ Chromosome-bound RNA polymerases are also capable of acting as barriers to cohesin translocation both in human and yeast.^{20,21} Moreover, transcribing RNA polymerases are not stationary; rather, they translocate and relocalize cohesin, which generates characteristic patterns of spatial organization around active genes.²⁰ Transcription

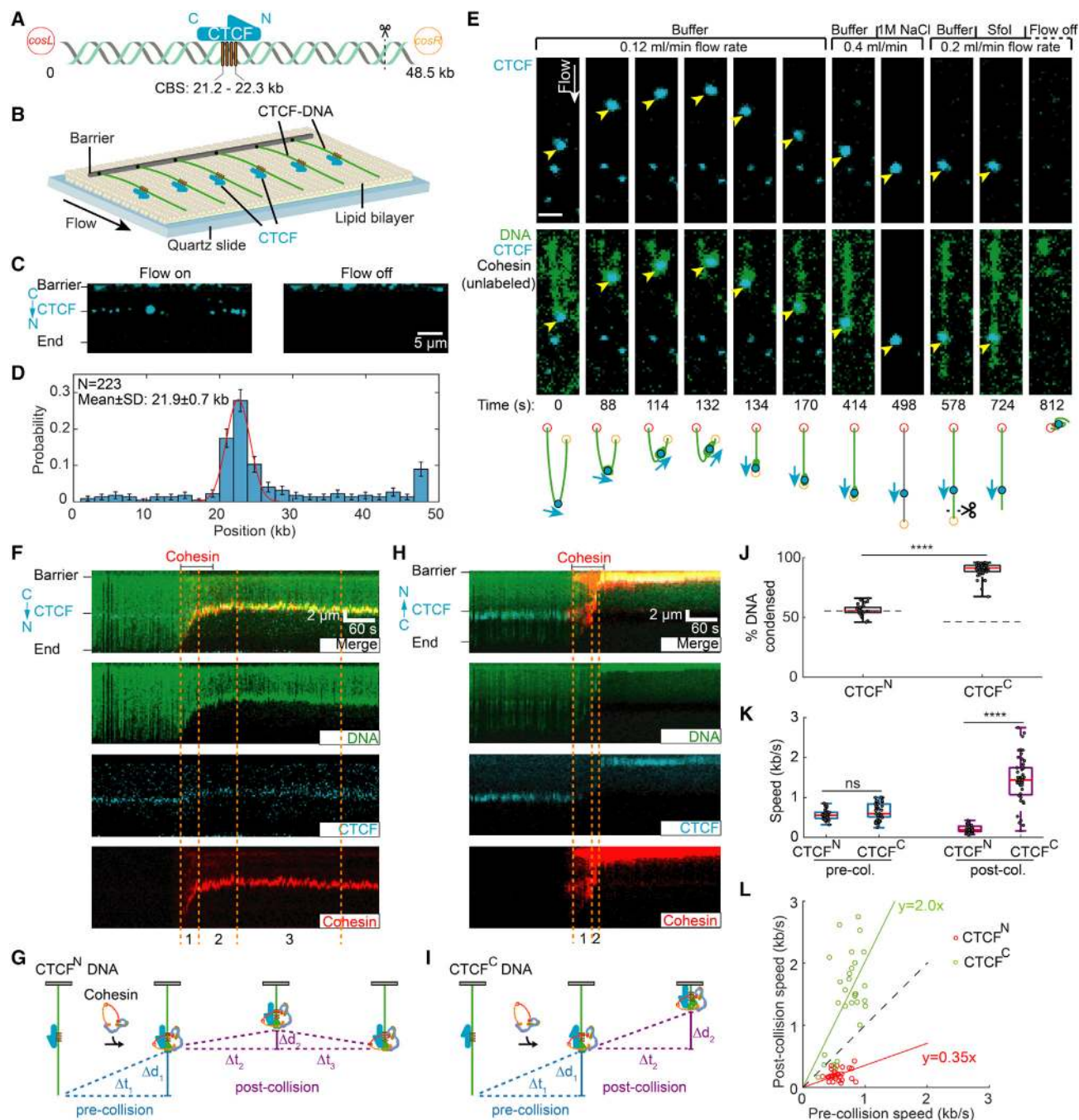


Figure 1. CTCF is a polar boundary for cohesin translocation

(A) Schematic of the DNA substrate. The location of the four CTCF-binding sites (CBSs) and the orientation of CTCF are shown in yellow boxes and a blue arrow, respectively. The black dashed line indicates the cutting site of restriction enzyme SfiI on DNA.

(B) An illustration of the DNA curtain assay where the *cosL* DNA end is anchored to the flowcell surface.

(C) Left: image showing Alexa 488-labeled CTCF binding to the DNA substrate. Right: turning off buffer flow retracts the DNA and CTCF to the barrier, confirming that CTCF is bound to the DNA.

(D) CTCF binding distribution on the DNA substrate. Red line: Gaussian fit. Error bars were generated by bootstrapping.

(E) Real-time visualization of CTCF stopping cohesin on U-shaped DNA. Both DNA ends are tethered to the flowcell surface. DNA is visualized with SYTOX Orange (green), and CTCF is labeled with an Alexa 488-conjugated antibody (blue). Upon cohesin injection, the DNA segment between the CTCF and right tether is compacted. At 134 s, the right tether detaches from the surface, causing the left DNA segment to extend by the buffer flow. A high-salt (1 M NaCl) wash at 498 s disrupts the looped DNA and washes out the SYTOX Orange stain. The DNA was restrained by reinjecting imaging buffer. To identify the *cosR* end, we injected the restriction enzyme SfiI, which cleaves near *cosR* at 724 s. Yellow arrows show the positions of CTCF. Scale bars: 3 μ m.

(legend continued on next page)

products, like RNA-DNA loops (R-loops), are also postulated to reinforce TADs.²² Whether R-loops can interact with cohesin and regulate loop extrusion is unknown. These studies all point to the intriguing possibility that not only CTCF, but also additional proteins and DNA structures, organize our 3D genomes.

Here, we use a combination of single-molecule studies and cryoelectron microscopy (cryo-EM) to show that CTCF and R-loops both block cohesin-mediated loop extrusion. CTCF-binding polarity controls cohesin-mediated DNA looping. Cohesin that encounters the non-permissive CTCF N terminus is blocked from further translocation and loop extrusion. The cryo-EM structure of the intact cohesin-CTCF complex reveals that this CTCF motif ahead of zinc fingers (ZFs) can only reach its binding site on the STAG1 cohesin subunit when the N terminus of CTCF faces cohesin. Remarkably, a C-terminally oriented CTCF accelerates cohesin translocation, causing increased DNA compaction. This suggests that CTCF shapes the 3D genome even when positioned in a permissive orientation relative to cohesin. DNA-bound Cas9 and Cas12a ribonucleoproteins (RNPs) are also polar cohesin barriers, indicating that cohesin stalling is intrinsic to this DNA motor and may be triggered by diverse proteins and/or DNA structures. Finally, we show that RNA-DNA hybrids (R-loops) are enriched with cohesin subunits *in vivo*. R-loops form insulating boundaries in the absence of CTCF and efficiently block cohesin-mediated DNA compaction *in vitro*. These results provide the first direct evidence that CTCF orientation and R-loops shape the 3D genome by directly regulating cohesin.

RESULTS

CTCF is a polar barrier to cohesin translocation on U-shaped DNA

We directly visualized cohesin-mediated looping and compaction of DNA bound with CTCF (Figure 1). CTCF assembles into clusters of 2–8 molecules on CBSs.^{32,33} We reconstituted this arrangement by inserting four co-directional CBSs into a 48.5 kb DNA substrate (Figure 1A; see STAR Methods).^{34,35} These CTCF motifs position the CTCF N terminus toward the right side of the DNA substrate, termed *cosR*. Full-length CTCF purified with a C-terminal maltose-binding protein (MBP)-FLAG tag forms a stable complex with double-stranded DNA (dsDNA) (Figures S1A and S1B). We fluorescently labeled CTCF with

Alexa 488-conjugated anti-FLAG antibodies. CTCF binding was visualized on aligned arrays of DNA molecules suspended above a lipid bilayer surface via total internal reflection fluorescence microscopy (Figures 1B and 1C; and see video data in STAR Methods).³⁶ Turning buffer flow off retracted both DNA and CTCF to the barrier, confirming that CTCF is bound to the DNA (Figure 1C). Nearly all CTCF molecules on DNA are bound to the CBSs (Figures 1C and 1D). The half-life of CTCF bound on the CBSs is 670 ± 60 s ($t_{1/2} \pm 95\%$ confidence interval [CI]; $n = 31$), which is ~ 5.2 -fold longer than its half-life on nonspecific DNA sites (Figure S1C). Fewer than four CBSs significantly reduced CTCF occupancy relative to non-specific DNA binding (Figure S1D). We estimate that the four CBSs bind 2 ± 1 (mean $\pm$ SD; $n = 233$) CTCF molecules, as indicated by the CTCF fluorescent intensity at the CBSs relative to the CTCF on nonspecific DNA (Figures S1E and S1F).

CTCF is a polar boundary for cohesin-mediated loop extrusion.^{10,11,25,37,38} Cohesin that encounters CTCF from its non-permissive, N-terminal side is proposed to stop extrusion. Whether encounters from the permissive, C-terminal side of CTCF (CTCF^C) can regulate cohesin is unknown. To determine how CTCF regulates cohesin, we directly observed loop extrusion on U-shaped CTCF-DNA. In these assays, both ends of the DNA substrate are biotinylated and tethered to the flowcell surface.^{36,39} DNA is visualized via the intercalating dye SYTOX Orange. Alexa 488-labeled CTCF is injected into the flowcell before unlabeled cohesin-Nipped-B-like (NIPBL)^C (hereafter referred to as cohesin in all single-molecule experiments) in the imaging buffer (Figures 1E and S2; see video data in STAR Methods). After cohesin is added, a representative DNA molecule shows gradual compaction of its right arm, indicating cohesin-mediated DNA looping (Figure 1E).^{36,40} Notably, the left arm of DNA is not compacted completely, suggesting that CTCF acts as a polar boundary to arrest cohesin. The right end of the molecule detached at 134 s, resulting in the linearization of the looped DNA molecule. While the left arm of the DNA was gradually extended, the right arm of the DNA stayed looped. This confirms cohesin-mediated looping of the right arm of the U-shaped DNA.

Since both ends of the U-shaped DNA substrate are biotinylated, these molecules are tethered with a random CBS orientation relative to the direction of cohesin translocation. We used *in situ* optical restriction enzyme mapping to determine the polarity of the CBSs. We first washed off cohesin by injecting

(F) Representative 3-color kymograph showing that CTCF (labeled with Alexa 488) arrests cohesin (labeled with Alexa 647) in the non-permissive (CTCF^N) orientation. Dashed lines indicate the pre- and post-collision time points depicted in (G).

(G) A schematic of cohesin-mediated compaction on a non-permissive CTCF-containing DNA and its analysis. Pre-collision: DNA is first condensed a distance Δd_1 for Δt_1 s. Post-collision: the DNA is further compacted (Δd_2) for a short time (Δt_2). The small DNA loop generated during Δt_2 is eventually dissipated (Δt_3), as seen by the CTCF/cohesin complex returning to the pre-collision position.

(H) Representative 3-color kymograph showing that CTCF permits further compaction after cohesin encounters at the permissive (CTCF^C) orientation. Dashed lines indicate the pre- and post-collision outcomes depicted in (I).

(I) A schematic of cohesin-mediated compaction on a permissive CTCF-containing DNA and its analysis for the pre-/post-collision. DNA continues to be compacted a distance Δd_2 for Δt_2 s after the collision.

(J) Quantification of the percentage of CTCF^N-DNA and CTCF^C-DNA condensed by cohesin. At least 32 DNA molecules were measured for each condition. The dashed lines indicate the CTCF-binding positions on DNA substrates.

(K) DNA compaction speed for the pre- and post-collisions with CTCF^N and CTCF^C. Boxplots indicate the median and quartiles. p values are obtained from two-tailed t test: ****p < 0.0001; ns, not significant.

(L) A comparison of the speed of individual cohesins before and after colliding with CTCF^N (red) or CTCF^C (green). The dashed line with a slope of 1 is included for reference.

a high-salt buffer at an increased flow rate (Figure 1E). After this stringent wash, CTCF remained bound at the target site, but DNA loop was disrupted, and DNA was re-extended. The restriction enzyme SfoI cuts at a single site 2.8 kb away from the *cosR* end (Figures 1A and S2A). When injected into the flowcell, SfoI cut the DNA molecule at what was formerly the right arm, indicating that this was the *cosR* side of the DNA substrate. Thus, cohesin compacted the *cosR*-proximal DNA arm and the encounter of cohesin with the CTCF N terminus arrested loop extrusion of the left DNA arm (Figures 1E, S2B, and S2C, blue arrows). Consistent with the random tethering of U-shaped DNAs, we observed that 58% ($n = 42/72$) of the molecules underwent complete compaction in both arms, as would be expected for a permissive CTCF-cohesin encounter (Figure S2D). The salt wash in Figure 1E indicates that cohesin remained bound to the CBS throughout DNA compaction. Furthermore, additional experiments on U-shaped DNA showed examples where cohesin compacted both arms of the U-shaped DNA without displacing CTCF from the CBS (Figure S2E). We also did not see CTCF displaced by cohesin from the CBS in any of the single-molecule experiments ($n = 67$). Overall, these experiments show that cohesin bypasses CTCF in the permissive orientation, but cannot displace CTCF from the CBS. These results suggest that CTCF acts as a boundary for cohesin-mediated DNA looping when the N terminus of CTCF is oriented toward cohesin.

CTCF can either block or accelerate cohesin translocation

We next used a three-color single-tethered DNA curtain assay to directly visualize how CTCF regulates cohesin translocation (Figure 1F). The DNA was tethered to the flowcell via a streptavidin-biotin linkage on either the *cosL* side (termed CTCF^N-DNA) or the *cosR* side (CTCF^C-DNA). The DNA, CTCF, and cohesin (via its STAG1 subunit) were labeled with different fluorophores that could be simultaneously imaged. Consistent with our prior observations, cohesin loads near the free DNA end and rapidly compacts the substrate.³⁶ Upon colliding with the N-terminal side of CTCF (CTCF^N), cohesin slowed drastically and translocated a few kb upstream of the CBS. The CTCF-cohesin complex then returned to the CBS, likely via force-induced dissipation of the DNA loop (Figures 1F and 1G; see video data in STAR Methods). All cohesin molecules stopped translocating after encountering CTCF^N ($n = 32$) (Figure 1J).

Collisions of cohesin with the CTCF^C produced drastically different results. Collisions with CTCF^C accelerated cohesin and compacted the entire DNA molecule ($n = 51$) (Figures 1I and 1J; see video data in STAR Methods). We compared cohesin translocation speeds before and after CTCF collisions in both orientations (Figure 1K). Before colliding with CTCF, cohesin speeds were indistinguishable in either CBS orientation (mean $\pm$ SD: 0.56 ± 0.13 kb s⁻¹ for cohesin-CTCF^N; 0.61 ± 0.21 kb s⁻¹ for cohesin-CTCF^C; $N > 32$ for both orientations). Non-permissive (CTCF^N) collisions slowed cohesin to a velocity of 0.2 ± 0.09 bp s⁻¹. Strikingly, permissive (CTCF^C) collisions increased the cohesin velocity to 1.42 ± 0.70 bp s⁻¹. This trend was also observed for changes in the velocity of individual molecules: non-permissive collisions slowed cohesin ~ 3 -fold whereas permissive collisions accelerated it by ~ 2 -fold (Fig-

ure 1L). We confirmed that wild-type (WT) CTCF that was not fluorescently labeled also blocked cohesin from the N-terminal side, indicating that this behavior is not induced by the fluorescent label (Figures S1G–S1J). Therefore, CTCF can arrest cohesin translocation when its N terminus is oriented toward cohesin. With its C terminus facing cohesin, CTCF accelerates cohesin after collision, possibly to reinforce domain boundaries.

Polar cohesin arrest requires the unstructured CTCF N-terminal domain

CTCF physically interacts with the STAG2-RAD21 cohesin subcomplex through its conserved N-terminal YDF motif.^{19,41,42} Cells with CTCF(Y226A/F228A) have fewer loops and weaker domain boundaries than WT CTCF.¹⁹ To determine whether this CTCF-cohesin interaction was required for blocking cohesin, we first characterized the ability of CTCF to arrest recombinant cohesin with STAG1(W337A/F347A; termed cohesin-WFA), which is deficient in binding the CTCF YDF motif¹⁹ (Figures 2A–2E). Cohesin-WFA did not stop after colliding with either CTCF^N or CTCF^C, and compacted DNA in both orientations ($n = 66$ and 30 for CTCF^N and CTCF^C, respectively). The pre- and post-collision velocities were also indistinguishable in either orientation (Figures 2D and 2E). We next tested CTCF(Y226A/F228A; CTCF-YFA) and a truncation mutant that only includes the 11 ZFs (CTCF-ZF) (Figure 2F). All CTCF mutants retained a high affinity for the CBS (Figures S3A and S3B). Strikingly, both CTCF-YFA (Figures S3C–S3G) and CTCF-ZF (Figures S3H–S3L) lost their functions as polar barriers of cohesin translocation. Thus, the interaction between STAG1 and the CTCF YDF motif is required for polar cohesin blockade.

CTCF-YFA and CTCF-ZF both reduced cohesin's speed and DNA compaction in an orientation-independent manner (Figure S3). This observation suggests that other regions of CTCF, including the ZFs, can partially block cohesin. We thus quantified the physical interactions between cohesin and CTCF mutants. To capture potential interactions between cohesin and CTCF mutants without DNA compaction, we increased the applied laminar force to ~ 0.7 pN. At this force, cohesin remains bound to the DNA but cannot translocate on it.³⁶ The vast majority of WT CTCF^N foci co-localized with cohesin ($n = 338/380$ molecules) (Figure 2G). This co-localization pattern was identical without ATP and with the ATP hydrolysis-deficient cohesin SMC1A(E1157Q)/SMC3(E1144Q) mutant (cohesin-EQ). In contrast, both CTCF^N-YFA and CTCF^N-ZF co-localized with less than $\sim 50\%$ of WT cohesin molecules ($n = 150/333$ and $195/397$ for CTCF^N-YFA and CTCF^N-ZF, respectively) (Figure 2G). Only 47% ($n = 105/224$) of the WT CTCF^N foci retained cohesin-WFA. Switching the DNA orientation to CTCF^C resulted in a similar co-localization defect (Figure S3M). Thus, the CTCF YDF motif is required for strong cohesin-CTCF binding. However, CTCF also physically interacts with cohesin via an internal region.

Structure of the cohesin-NIPBL^C-CTCF-DNA complex

To understand the mechanism by which CTCF blocks cohesin in an orientation-specific manner, we solved the structure of cohesin-NIPBL^C in complex with CTCF using cryo-EM. The complex was reconstituted on a 118-base pair (bp) dsDNA that included a 41-bp CBS at one end. Cohesin-NIPBL^C remained associated

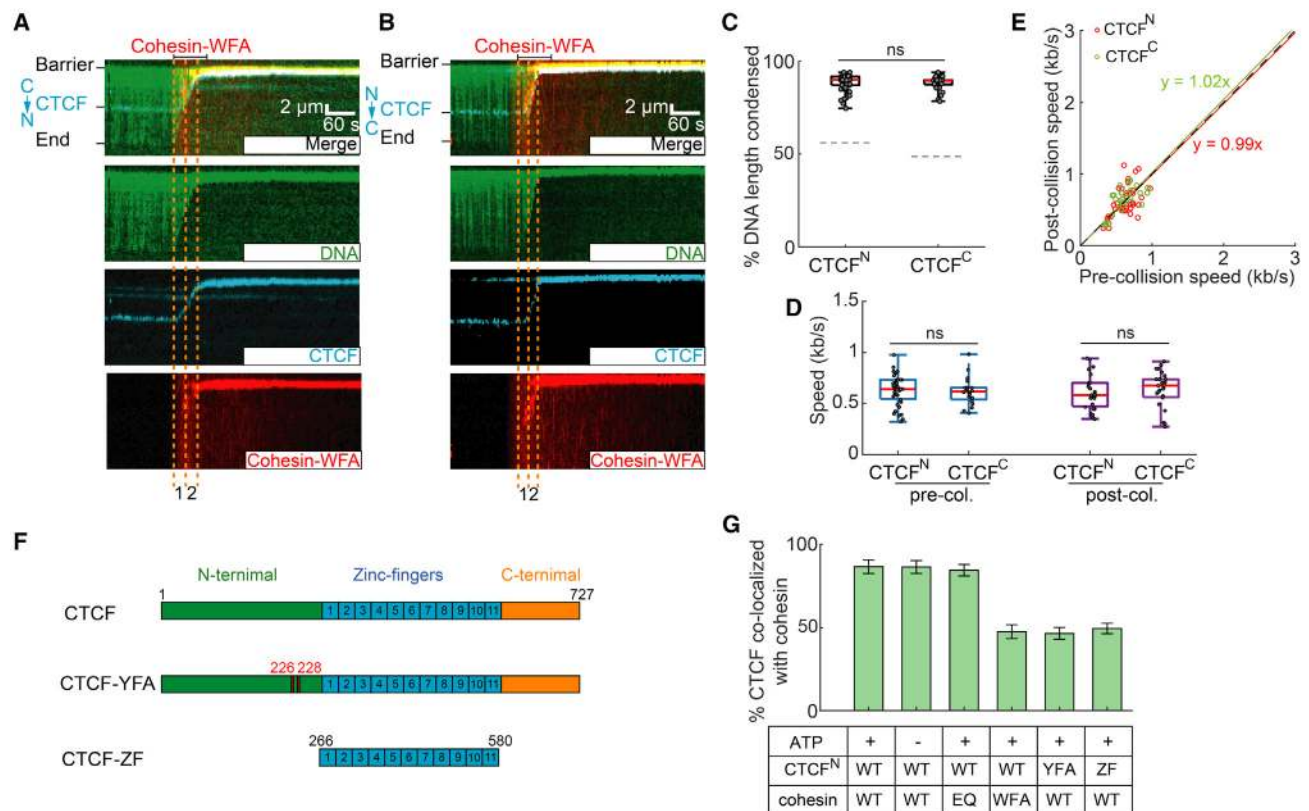


Figure 2. An interaction between STAG1 and the CTCF N-terminal region are essential for polar cohesin arrest

(A and B) Representative kymographs showing that cohesin-STAG1(W337A/F347A), termed cohesin-WFA, can completely compact DNA pre-bound with (A) CTCF^N and (B) CTCF^C.
 (C) Quantification of the CTCF^N-DNA and CTCF^C-DNA condensed by cohesin-WFA. The dashed lines indicate the CTCF-binding positions on DNA substrates.
 (D) Cohesin-WFA speed pre- and post-collisions with CTCF^N or CTCF^C.
 (E) Correlation between the speeds of individual cohesins before and after colliding with CTCF^N (red) or CTCF^C (green). The dashed line is a guide with a slope of 1.
 (F) Schematic of wild-type CTCF, CTCF Y226A/F228A mutant (CTCF-YFA), and the zinc-finger truncation (CTCF-ZF).
 (G) Percent of CTCF or its mutants co-localized with cohesin variants on CTCF^N-DNA. At least 30 DNA molecules were measured for each experiment. p values are obtained from two-tailed t test; ns, not significant.

with CTCF-bound dsDNA in the presence of ADP•BeF³⁻ (Figures S4A and S4B). We further stabilized this complex via mild cross-linking with BS³ before sucrose gradient ultracentrifugation and single-particle cryo-EM analysis.

3D classification and refinement generated three cryo-EM maps of the complex in distinct conformations, only one of which contained one CTCF molecule bound to the cohesin-NIPBL-DNA complex (Figures S4C–S4I; Table S1). The map of this conformation had an overall resolution of 6.5 Å, which allowed unambiguous rigid-body docking of the models of cohesin-NIPBL and CTCF ZFs with DNA to produce the structure of the cohesin-NIPBL-CTCF-DNA complex^{43–45} (Figures 3A and S4J). The CTCF N and C termini are predicted to be unstructured, and accordingly, are invisible in the map. Previous crystal structure showed that the YDF motif of CTCF interacts with STAG2 and RAD21.¹⁹ To visualize the density of YDF motif in our cryo-EM structure, we performed focused refinement on the STAG1-RAD21-CTCF-DNA part in three maps of the complex structures (Figures S4M–S4O). The density corresponding to the YDF motif is clearly present in two of the three maps of the complex (Figures S4M and

S4N). Extra density that corresponds to the YDF motif is also present on the surface of STAG1-RAD21 in the map for cohesin-NIPBL-CTCF-DNA (Figure S4O), albeit the map is not as good as in the other two maps due to the limited resolution.

In the complex, one end of the DNA molecule is captured by SMC1-SMC3 heterodimer and NIPBL, similar to the cohesin-NIPBL-DNA complex without CTCF (Figure 3A).^{43,46,47} The middle region of DNA is bound by STAG1 (Figure 3B). Previous studies have shown that the huntingtin, elongation factor 3, a subunit of protein phosphatase 2A and TOR1 (HEAT) repeat proteins of SMC complexes, including STAG1/2 in cohesin, NIPBL, and in chromosome-associated proteins (CAP) CAP-D and CAP-G (Ycg1 in yeast) in condensin, participate in DNA binding. Unlike NIPBL that contacts DNA via its left and right arms on one side of the U structure, STAG1 binds to DNA through both the bottom of the left arm and the tops of both arms (Figures 3C and 3D).⁴³ These DNA recognition regions in STAG1 are enriched in positively charged residues (Figure 3B). DNA traverses between the tops of the U-shaped STAG1, which is similar to the DNA-binding mode by yeast condensin subunit Ycg1 (Figure 3E).^{48,49} However, STAG1

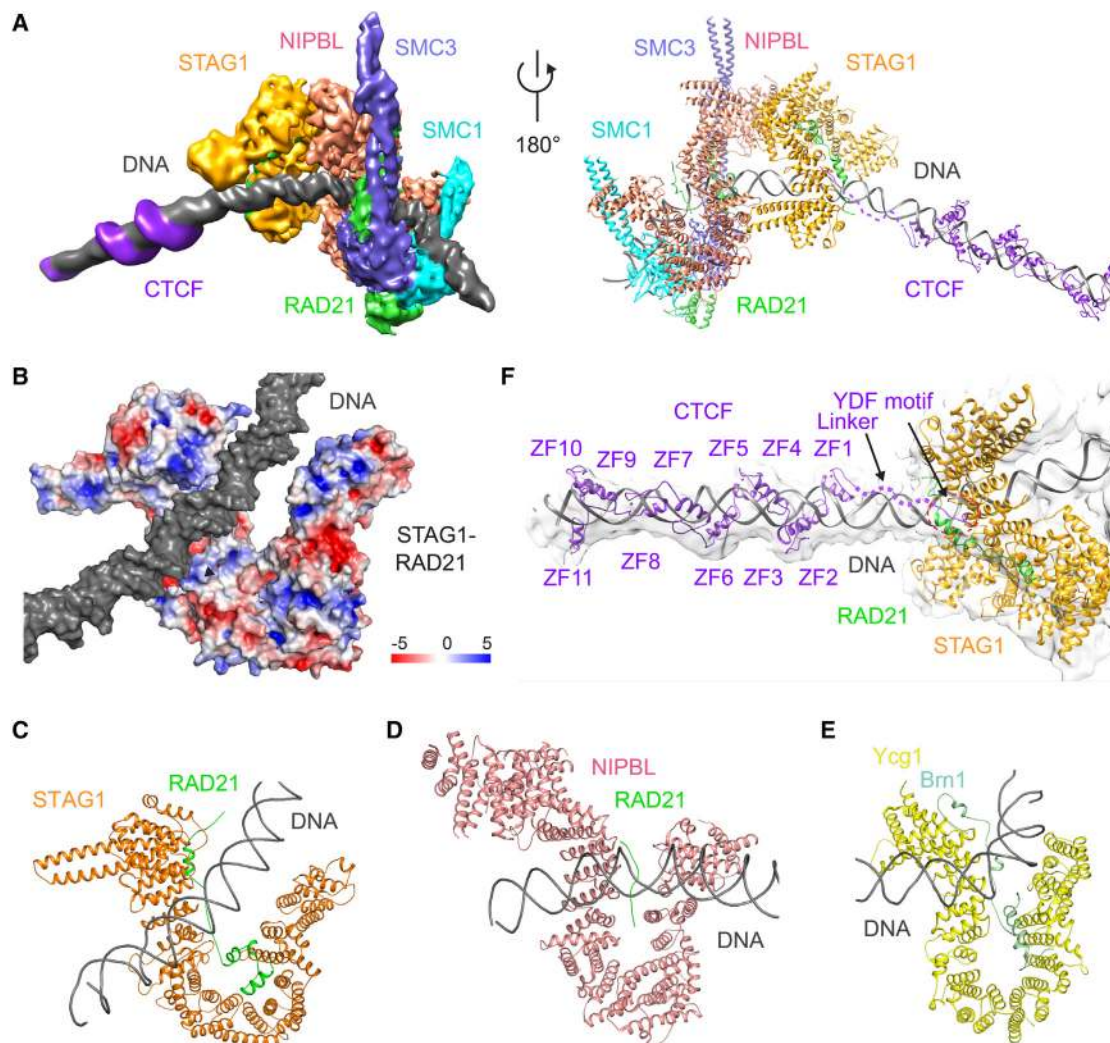


Figure 3. Structure of the human cohesin-NIPBL-CTCF-DNA complex

(A) Cryo-EM map (left) and model (right) of human cohesin-NIPBL-CTCF-DNA complex. DNA is captured by cohesin and NIPBL at one end and by CTCF at the other end, while its middle region contacts the top of both sides of U-shaped STAG1.

(B) Surface electrostatic potential of STAG1-RAD21 subcomplex. DNA contacts positively charged regions in STAG1 and RAD21.

(C–E) Structural comparison of HEAT repeat proteins STAG1 (C), NIPBL (D), and Ycg1 (E) binding to DNA duplex.

(F) Locally refined map of the STAG1-CTCF-DNA subcomplex. The models of STAG1, STAG1-bound RAD21 region, DNA, and CTCF YDF motif and ZFs are shown. The CTCF linker region flanked by the YDF motif and ZFs contacts DNA.

possesses a wider central cleft than Ycg1, resulting in a relatively loose binding of STAG1 to DNA, which might be an intrinsic property of cohesin. It is also possible that other unidentified factors can strengthen STAG1-DNA interactions at domain boundaries.

CTCF binds to the CBS at the other end of the DNA molecule, with its N terminus pointing toward cohesin. The structure thus captures the extrusion-blocking collision complex of cohesin-CTCF. The conserved YDF motif of CTCF binds to the previously characterized site on the STAG1-RAD21 subcomplex¹⁹ (Figure 3F). In addition to the N terminus, zinc finger 1 (ZF1) of CTCF contributes to cohesin positioning at CBSs, boundary insulation, and loop formation.^{41,42,50,51} However, cohesin does not directly contact CTCF ZFs (Figures 3A and 3F). Thus, CTCF-ZF1 regulates cohesin via an indirect mechanism.

The YDF motif that directly binds STAG1 is conserved in CTCF proteins of various species, including *Drosophila* (Figure S5A). Yet, CTCF is not enriched at TAD boundaries and loop anchors in *Drosophila*,^{52–54} suggesting that *Drosophila* CTCF cannot block cohesin. A sequence alignment of vertebrate CTCFs shows that the N-terminal region contains a patch of lysine residues close to ZF1 that may interact with DNA (Figure S5A). We also observed additional weak density adjacent to human CTCF ZFs on the surface of DNA in the locally refined maps (Figure 3F), indicating that this basic linker binds to DNA and may be important for blocking DNA compaction by cohesin. Interestingly, this basic linker is missing in *Drosophila* CTCF (Figure S5A), which could provide a possible explanation for the inability of *Drosophila* CTCF to stop cohesin. To demonstrate this possibility, we purified a

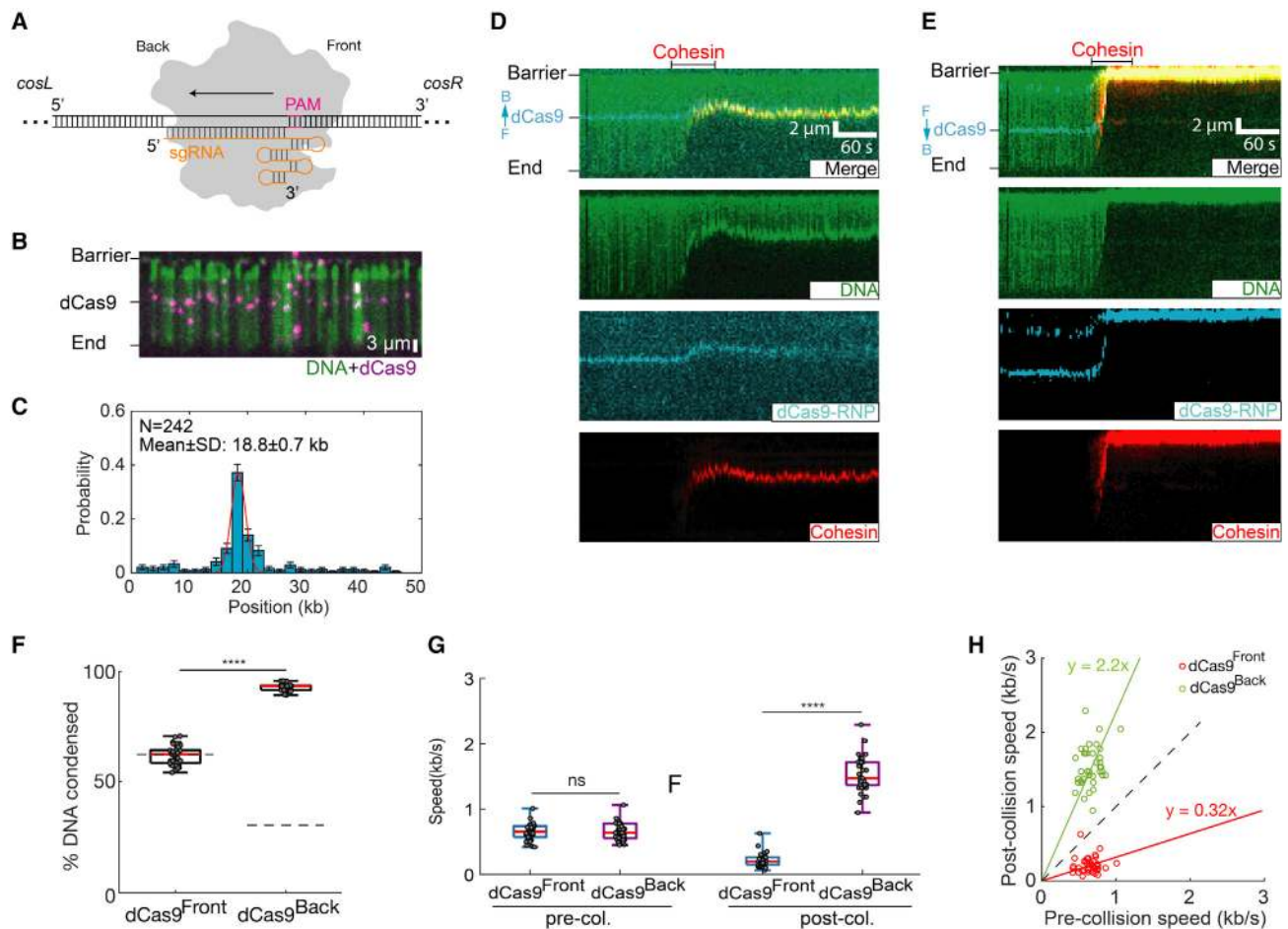


Figure 4. Cas9 is a polar cohesin barrier

(A) Schematic of Cas9 binding its target DNA site. sgRNA is in orange. The direction of R-loop formation is indicated with an arrow. The Cas9 protospacer adjacent motif (PAM) faces the *cosR* DNA end, termed dCas9^{Front}; PAM-distal side is termed dCas9^{Back}.
 (B) Image of Alexa 488-labeled dCas9 binding its target DNA.
 (C) Binding distribution of dCas9 on the DNA substrate. Red line: Gaussian fit.
 (D) Representative kymographs showing that dCas9 blocks cohesin when cohesin collides with the PAM-proximal dCas9 face (dCas9^{Front}). For these experiments, the DNA is tethered via its *cosL* end. F, front; B, back.
 (E) When cohesin collides with the PAM-distal dCas9 face (dCas9^{Back}), its post-collision speed increases.
 (F) Quantification of the percentage of dCas9^{Front}-DNA and dCas9^{Back}-DNA condensed by cohesin (N > 40 for each condition).
 (G) Comparison of the pre- and post-collision cohesin speeds for dCas9^{Front} and dCas9^{Back}. p values are obtained from two-tailed t test: ****p < 0.0001; ns, not significant.
 (H) A scatterplot showing the relationship for individual cohesin speed before and after collision with dCas9^{Front} (red) and dCas9^{Back} (green). The dashed line is for a reference (slope = 1).

chimeric CTCF with residues 261–264 replaced by the corresponding *Drosophila melanogaster* linker (CTCF-DM[251–264]) (Figure S5B). CTCF-DM(251–264) cannot block cohesin translocation compared with WT human CTCF (Figures S5C and S5D), indicating that the linker constitutes a key functional difference between human and fly CTCFs.

Polar arrest of cohesin by Cas9 and Cas12a ribonucleoproteins

To further probe the mechanism of cohesin arrest, we used *S. pyogenes* Cas9 as a model roadblock with a defined polarity. Nuclease-dead Cas9 (dCas9) RNP was reconstituted by mixing

3xFLAG-dCas9 with a single-guide RNA (sgRNA). The protospacer adjacent motif (PAM) of the sgRNA faces the *cosR* end of the target DNA strand (Figure 4A). As expected, the dCas9 RNP labeled with a fluorescent Alexa 488-anti-FLAG antibody bound to its DNA target (Figures 4B and 4C).⁵⁵ When the DNA is tethered via its *cosL* end, cohesin collides with dCas9 from its PAM-proximal side (named dCas9^{Front}) and when the DNA is tethered via its *cosR* end, cohesin encounters the PAM-distal side (dCas9^{Back}; see Figure 4A).

Cohesin compacts the DNA until it encounters dCas9 in either orientation. Strikingly, we observed different behaviors with dCas9^{Front} and dCas9^{Back} (Figures 4D and 4E; see video data

in STAR Methods). dCas9^{Front} slowed cohesin ~ 3 -fold relative to its pre-collision speed (0.61 ± 0.12 kb/s; $n = 40$) and eventually arrested cohesin at the collision site (Figures 4F and 4G). In contrast, collisions with dCas9^{Back} increased cohesin's speed ~ 2.2 -fold (1.52 ± 0.27 kb/s; $n = 41$) and led to nearly complete DNA compaction (Figures 4F–4H). Thus, dCas9 recapitulates the polar cohesin arrest and acceleration that we observed with CTCF (Figure 1). Our surprising finding that dCas9 arrests cohesin in a polar fashion explains a recent *in vitro* report that gold nanoparticle attached to dCas9-RNP only blocks $\sim 50\%$ of cohesins.⁵⁶ This partial effect is likely due to the unresolved collision polarity in that study. More importantly, our results also explain how dCas9 establishes TADs in mammalian cells.⁵⁷

We next observed cohesin's collisions with the nuclease-inactive *Acidaminococcus* sp. Cas12a (dCas12a) RNP. Cas12a and Cas9 are structurally and biochemically divergent RNA-guided nucleases. dCas12a recognizes its PAM on the 3' side of the target DNA strand (the PAM-proximal side is named dCas12a^{Front}; the PAM-distal side is named dCas12a^{Back}; Figure S5E), which is opposite to dCas9 (Figure 4A). Fluorescently labeled dCas12a efficiently bound its target site (Figures S5F and S5G).⁵⁸ dCas12a slowed and eventually stopped cohesin but only when cohesin approached from the dCas12a^{Front} side (Figures S5H–S5L). Collisions with dCas12a^{Back} accelerated cohesin ~ 1.8 -fold ($n = 53$). Similar to CTCF, the fluorescent label did not affect cohesin translocation in these assays (Figures S5M–S5R). To investigate whether a specific interaction in STAG1 is required for polar arrest by CRISPR nucleases, we performed additional experiments with cohesin-WFA (Figures S5S–S5X). As with WT complexes, cohesin-WFA is also arrested by dCas9^{Front} but not by dCas9^{Back} (Figures S5S–S5U). Similar results were observed for dCas12a (Figures S5V–S5X).

Together, although the mechanisms are different, CTCF, dCas9, and dCas12a can all arrest or accelerate cohesin, depending on the polarity of the encounter. Polar arrest is a general feature of cohesin's translocation cycle that can be elicited by diverse roadblocks.

R-loops act as barriers to cohesin translocation

Target-bound Cas9 and Cas12a both form an RNA:DNA hybrid (R-loop) with a displaced single-stranded DNA. R-loops also form genome-wide during transcription via hybridization of the nascent transcript with the template DNA strand. Cohesin binds RNA via its STAG1/2 subunits *in vitro*,⁵⁹ and STAG1/2 proteins are enriched at R-loops in cells.⁶⁰ Moreover, apoCas9 doesn't block cohesin translocation, suggesting that R-loops may impede cohesin directly.

We generated stable R-loops *in vitro* and observed their impact on cohesin translocation (Figure 5; see video data in STAR Methods). R-loops were assembled via concatemerization of a plasmid encoding the mouse *Airm* gene, followed by *in vitro* transcription and RNase A treatment (Figure S6A).^{59,61} R-loops from the *Airm* gene are stable both in cells and *in vitro*.^{59,61} The DNA concatemers varied in length from 17 to 110 kb (Figure S6B). Transcription did not appreciably change the distribution of DNA lengths (Figure S6C). We estimate 3 ± 2 R-loops per DNA molecule by fluorescently imaging these structures with the S9.6 antibody conjugated with Alexa 488

(Figures 5B and S6D). R-loops were separated by multiples of 4 kb, as expected for a DNA substrate that is generated via multicopy ligation of the same 4 kb long plasmid (Figure S6E). These DNA molecules were biotinylated and injected into the flowcell for single-molecule imaging. About 44% of the R-loops bound cohesin, confirming a physical interaction, likely with STAG1 (Figures 5C and S6F).⁵⁹ Cohesin did not fully compact DNA in the presence of R-loops and slowed 0.7-fold (0.42 ± 0.31 kb/s; $N = 37$) (Figures 5D–5F). In contrast, cohesin completely compacted non-transcribed DNA or transcribed substrates that had been digested with RNase H to remove the R-loops (Figures 5D, S6G, and S6H). As expected, R-loop substrates that were pre-treated with RNase H did not slow cohesin (Figure 5E). We ruled out that the S9.6 antibody caused cohesin to stall by first imaging the R-loop collisions and then labeling R-loops after the experiment was complete (Figures S6I and S6J). We obtained similar results with both R-loop labeling approaches, indicating that R-loops indeed slow cohesin on their own. Cohesin stalled 19 ± 8 kb after colliding with the first R-loop ($n = 45$), indicating that a single encounter was insufficient to halt translocation. We estimate that complete cohesin arrest required 5 ± 2 R-loop collisions on average (Figures 5G and S6K). A stringent 1 M NaCl wash re-extended partially looped DNA molecules, but did not disrupt the tight cohesin-R-loop interaction (Figure 5H). In contrast, 1M NaCl is sufficient to remove cohesin from naked DNA. Although we could not distinguish the direction of cohesin and R-loop collisions in these assays, our results indicate that multiple R-loops are sufficient to stall cohesin translocation *in vitro*, even in the absence of all transcription machinery.

R-loops are enriched for cohesin complexes and insulate genomic contacts in cells

To test whether R-loops also act as cohesin barriers in cells, we analyzed published chromatin immunoprecipitation sequencing (ChIP-seq) and DNA-RNA immunoprecipitation sequencing (DRIP-seq) datasets in mouse embryonic fibroblasts (MEFs)^{62,63} (Table S4). We analyzed chromatin localization patterns for the cohesin subunits Rad21 and Stag1 as proxies for the entire cohesin complex (Figure 6; Table S4). Peak overlaps were determined using the BEDTools suite with default parameters,⁶⁴ and they were deemed significant by BEDTools fisher, ChIPseeker and Genomic HyperBrowser (see STAR Methods; Table S5). Fifteen percent of cohesin subunit peaks overlap with R-loop peaks in WT MEFs (Figure 6A; Table S5). Overlaps between R-loops and Rad21 or Stag1 were nearly identical, indicating that these signals likely represent a complete cohesin complex. Knocking out the cohesin-release factor Wapl did not change the cohesin and R-loop overlap genome-wide. However, the cohesin-R-loop overlap increases to $\sim 26\%$ in CTCF-depleted cells and CTCF/Wapl double-knockout (DKO) cells, suggesting that more cohesins interact with R-loops when CTCF is ablated. Cohesin and R-loop overlap is enriched at promoter and intronic regions, consistent with the pervasive presence of R-loops between the transcription start site (TSS) and the first exon-intron junction.⁶⁵ This enrichment is significantly enhanced in CTCF or CTCF/Wapl KO cell lines, indicating that R-loops provide a secondary signal for 3D genome organization

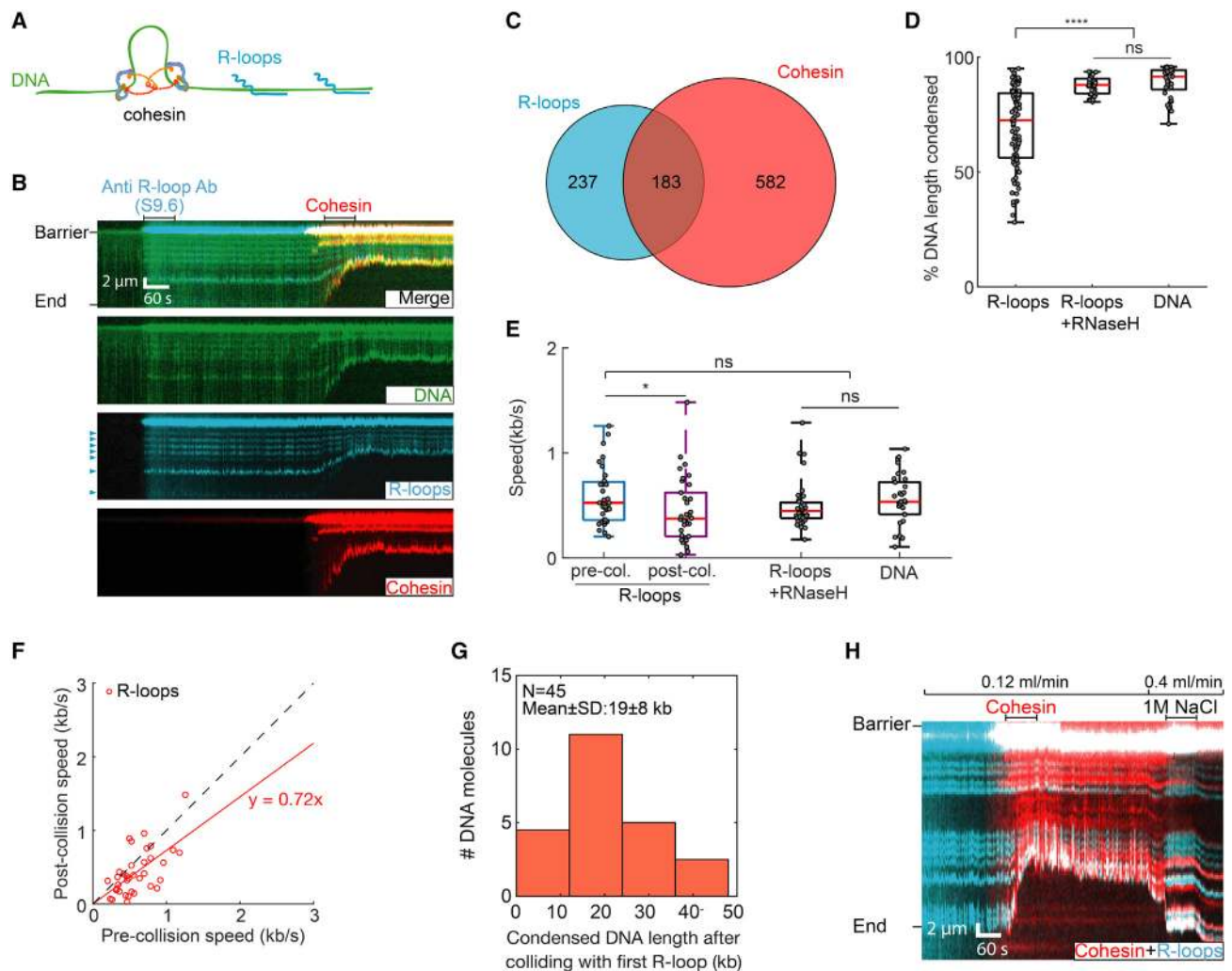


Figure 5. R-loops interact with cohesin and slow its translocation

(A) Schematic of cohesin translocation on the R-loops DNA substrate.

(B) Representative kymographs showing cohesin colliding with R-loops. An Alexa 488-conjugated S9.6 antibody is used to image the R-loops prior to cohesin injection. R-loops are indicated by arrows.

(C) Venn diagram showing co-localization of R-loops and cohesin ($n = 110$ DNA molecules).

(D) R-loops significantly decrease DNA compaction, as compared with R-loops pre-treated with RNase H and non-transcribed DNA. $N > 38$ DNA molecules for each condition.

(E) After colliding with an R-loop, cohesin slows its DNA compaction. $N > 38$ for all conditions. p values are obtained from two-tailed t test: $*p < 0.05$; ns, not significant.

(F) Individual cohesin molecules slow upon colliding with their first R-loop. Dashed line is a guide with a slope of 1.

(G) The counts of DNA molecules showing cohesin continues to compact DNA for ~ 20 kb after colliding with the first R-loop.

(H) Kymograph showing that a high-salt (1 M NaCl) wash disrupts the compacted DNA. However, cohesin remains associated with the R-loop.

(Figure 6B). To further confirm the significance of the overlap between cohesin and R-loop peaks, we mapped R-loop prevalence in a 5 kb region upstream/downstream of the cohesin-R-loop overlapped region. R-loops are prevalent around the cohesin-R-loop overlap region (Figure 6C) but not a few kb away from the overlaps. We also repeated this bioinformatic analysis for WT HeLa and K562 human cell lines, where both DRIP-seq and cohesin ChIP-seq datasets are publicly available.^{63,66–68} Human cell lines also showed strong cohesin enrichment near R-loops, with the strongest enrichment at promoters and intron

regions (Figures S6L–S6N; Tables S6 and S7). We conclude that cohesin and R-loops occupy the same genomic sites, and that R-loops are additional barriers for cohesin translocation in cells.

Next, we analyzed genomic contacts in the vicinity of R-loops using publicly available high-throughput chromosome conformation capture (Hi-C)²⁰ and DRIP-seq datasets in MEFs.⁶³ A map of chromosome contact enrichment averaged over and centered on oriented R-loops indicates that R-loops act as insulators for upstream and downstream contacts in WT MEFs

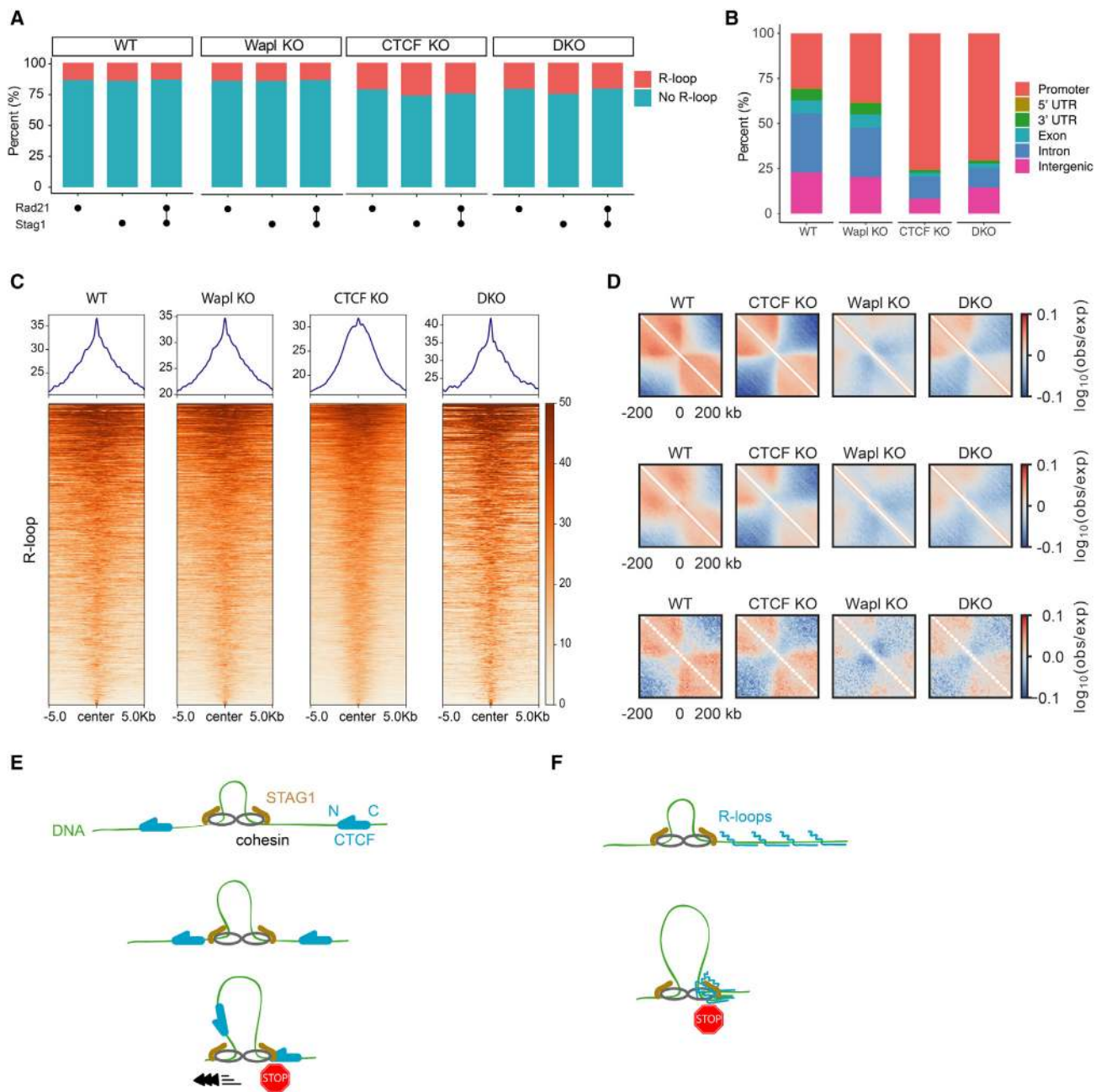


Figure 6. R-loops act as barriers to cohesin-mediated loop extrusion in cells

(A) Cohesin subunit Rad21 and Stag1 peak positions overlap with R-loops in WT, Wapl knockout (KO), CTCF KO, and CTCF/Wapl double KO (DKO) MEFs, as defined by ChIP-seq and DRIP-seq, respectively. Both previously published datasets were collected in mouse embryonic fibroblasts (MEFs).

(B) Genomic features of overlapping regions of Rad21, Stag1, and R-loop peaks in the indicated MEFs.

(C) Read-density profiles and heatmaps of R-loop reads across overlaps of Rad21, Stag1, and R-loop in the indicated MEFs.

(D) Average maps of chromosome contact enrichment ("observed-over-expected"; see [supplemental information](#)) in MEFs (WT and mutants) in the vicinity of all R-loops (top; $n = 39,680$; R-loops centered at 0 kb). To minimize effects of transcription start sites (TSSs) and RNA polymerase, we recomputed the maps excluding R-loops located within 10 kb of a transcription start site (middle; $n = 27,542$). Intergenic R-loops ($n = 5,392$) also generated insulation (bottom) in WT and mutant MEFs.

(E) A summary of cohesin regulation by CTCF. Cohesin is blocked by the N terminus of CTCF through its interaction with STAG1 but increases its velocity when it encounters the C terminus of CTCF.

(F) A summary of the effect of R-loop clusters on cohesin translocation.

(Figure 6D, top). Insulation at R-loops depends on cohesin since the enrichment of genomic contacts vanishes in Smc3 KO cells (Figure S6O). Knocking out CTCF somewhat increases the strength of the insulation, presumably because cohesin no longer accumulates at CTCF boundaries. When Wapl is depleted, R-loops are not insulating. This is likely because cohesin accumulates along the entire genome and has more time to traverse R-loops. A double CTCF and Wapl KO (DKO) partially restores insulation, further demonstrating that R-loops can help shape the 3D genome. Both R-loops and RNA polymerase are enriched at promoters, and RNA polymerase generates insulation through its interactions with cohesin,²⁰ partially confounding our analysis. To avoid the possible effects from non-R-loop factors in promoter regions, we piled up Hi-C maps centered on R-loops but excluded those that are found within 10 kb of a TSS (Figure 6D, middle). As an even more stringent analysis, we considered only intergenic regions while also excluding R-loops within 10 kb of a TSS (Figure 6D, bottom). These analyses confirm insulation near intergenic R-loops and R-loops away from TSSs, with a somewhat attenuated signal compared with the average over all R-loops. Taken together, the *in vitro* and *in vivo* analyses indicate that R-loops can act as barriers for cohesin translocation to shape the 3D genome.

DISCUSSION

Here, we provide direct evidence that CTCF is a polar barrier to cohesin-mediated DNA loop extrusion. Our structure of the cohesin-NIPBL-CTCF-DNA complex represents the extrusion-arrested state. In this state, cohesin adopts a fully folded conformation, with STAG1/2 engaging the N-terminal YDF motif of CTCF. This arrangement strongly suggests that as cohesin translocates on DNA, STAG1/2 is positioned in the front end of the complex. When cohesin approaches the N terminus of CTCF, the YDF motif in the N-terminal region of CTCF can interact with STAG1/2, thus blocking cohesin translocation (Figures 6E and S6P). Our structure thus explains the molecular basis of the polar cohesin arrest by CTCF.

Cohesin translocation on DNA is accelerated when it first encounters the C terminus of CTCF (Figure 6F). Such acceleration further improves cohesin's processivity and may be important for reinforcing domain boundaries in cells. This acceleration requires the interaction between STAG1/2 and the YDF motif of CTCF. However, the extrusion-arrested structure reported here suggests that the N-terminal CTCF linker, which directly contacts DNA region ahead of ZF-binding site, may not allow the YDF motif to reach STAG1/2 when cohesin approaches from the C terminus of CTCF (Figures S6Q and S6R). We speculate that CTCF might interact with STAG1/2 in an actively extruding cohesin-NIPBL complex and that this interaction further activates cohesin translocation. The molecular basis for the increased cohesin velocity is unclear. One possibility is that CTCF suppresses cohesin's tendency to slip on DNA, especially at higher applied forces.^{36,40} By directly interacting with STAG1/2, CTCF may act as a processivity factor that prevents microscopic cohesin slipping during its translocation cycle to reinforce cohesin's loop extrusion at high tensions in mammalian cells. Additional structural and biochemical studies are needed to fully

elucidate how cohesin extrudes loops and how CTCF reinforces this process.

Remarkably, both Cas9 and Cas12a RNPs can recapitulate polar cohesin arrest and acceleration, suggesting that CTCF is not unique in this regard. We only observed polar arrest/acceleration for the RNP but not the apo-Cas9/Cas12a complexes, suggesting that the protein-generated R-loop is important for this activity. Although cohesin has not evolved to interact with such RNPs *in vivo*, this result hints that *S. pombe*, *S. cerevisiae*, *C. elegans*, and *A. thaliana* may not need a CTCF homolog to organize their genomes. These organisms can form distinct chromatin domains reminiscent of TADs seen in humans.⁶⁹ Moreover, *Drosophila* CTCF performs fundamentally different functions from the human homolog, and its chromosome contact domains can form without stabilized point-to-point border interactions between CTCF sites.^{31,70} Thus, cohesin must recognize additional CTCF-independent signals to form TADs, and these may include unidentified DNA-binding proteins or nucleic-acid structures. Even in human cells, some TAD boundary elements are not CTCF-dependent, suggesting that additional principles can also establish chromosome contact domains.¹⁰

While this paper was in revision, another group used a similar single-molecule imaging approach to demonstrate that CTCF blocks loop-extruding cohesin in an orientation-dependent manner.⁷¹ Notably, they reported that CTCF is an active regulator of cohesin-mediated loop extrusion that can be modulated by DNA tension. They did not observe significant changes in the rate of loop extrusion before and after cohesin-CTCF collision and found that dCas9 weakly arrests cohesin, which is contrary to other reports *in vitro* and *in vivo*.^{56,57} The reasons for the discrepancies may be related to multiple differences between our studies. First, we co-purified the recombinant cohesin-NIPBL^C complex that was expressed in insect cells. In contrast, Davidson et al. used both HeLa and insect cells to express cohesin and NIPBL-MAU2 subcomplexes. In that study, cohesin and NIPBL-MAU2 were preincubated at various ratios right before injecting into the flowcell. We also used ~48 kb long DNA substrates with four CBSs, whereas the second study only had a single CBS on a much shorter DNA molecule. Finally, the experiments described herein are primarily from single-tethered DNA molecules, whereas the second study observed cohesin translocation on double-tethered DNA experiments. When both ends of the DNA are tethered, tension accumulates because of cohesin looping as opposed to the constant application of buffer flow. Additional studies will be required to dissect the mechanistic details that lead to these small discrepancies, as well as to the mechanisms of cohesin arrest by other roadblocks.

Here, we show that R-loops can arrest cohesin, and that cohesin is enriched at R-loops *in vivo*. Our *in vivo* co-localization and Hi-C analyses are correlative and cannot rule out an indirect mechanism for R-loop-associated cohesin and contact enrichment, possibly via RNA polymerase-mediated insulation at these sites. Future experiments will be required to directly test this hypothesis. Additional evidence for the importance of R-loops includes the formation of fine-scale chromatin loops connecting the promoter, the enhancer, and downstream exon regions soon after induction of transcription.⁷² Interestingly, RNase H1

destroys these formed loops and eliminates cohesin binding to these sites, suggesting that these chromatin loops depend on R-loops and cohesin. Moreover, accumulation of RNA-DNA hybrids flanking CBSs decreases CTCF binding to CBSs in DIS3-deficient B cells and disorganizes cohesin localization, negatively impacting the integrity of the TAD containing the immunoglobulin heavy-chain (Igh) locus.⁷³ Active transcription also limits cohesin-mediated loop extrusion during recombination-activating gene (RAG) scanning.⁵⁷ We conclude that R-loops can arrest cohesin-catalyzed DNA looping both *in vitro* and *in vivo* with broad implications for the roles of R-loops and other roadblocks in shaping 3D genome organization in cells.

Limitations of the study

The optical resolution of the fluorescence microscope used in these studies is constrained by the diffraction limit and the Brownian motion of the DNA and protein particles. This limits our ability to see the molecular details of how cohesin interacts with roadblocks with nanometer precision. The DNA molecules are stretched by the application of buffer flow (~0.2 pN of applied force). This extends the DNA such that it can be imaged but can affect cohesin and CTCF.⁷¹ In the cell, cohesin moves on a highly chromatinized DNA that also includes both R-loops, CTCF, and a plethora of other DNA-binding proteins and RNA molecules. Biochemical reconstitutions can only capture a small subset of these interactions in a purified system. We determined the structure of the cohesin complex stalled by CTCF on DNA. In our structures, STAG1 binding is highly flexible. Recent studies suggest that the acetyltransferase ESCO1 and the HEAT repeat protein PDS5 are required for the formation of loops and the maintenance of the convergent rules of CBSs.^{17,74,75} Future work will explore how these factors interact with cohesin and CTCF at TAD boundaries and thus contribute to loop formation.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.molcel.2023.07.006>.

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AUTHOR CONTRIBUTIONS

H.Y., X.-c.B., and I.J.F. co-supervised and designed the project. H.Z. designed and performed all single-molecule assays and analyzed single-molecule and DRIP/ChIP-seq data. Z.S. performed protein purification. Z.S. and X.-c.B. performed cryo-EM studies. Y.K. contributed to project design and protein purification. E.J.B. performed the Hi-C analysis. All authors contributed to the writing of the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

INCLUSION AND DIVERSITY

We support inclusive, diverse, and equitable conduct of research.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Goat anti-Mouse IgG (H+L), Superclonal Recombinant Secondary Antibody, Alexa Fluor 488	Invitrogen	Cat#A28175; RRID:AB_2536161
6xHis Monoclonal Antibody (Albumin Free)	Takara	Cat#631212; RRID:AB_2721905
F(ab') ₂ -Goat anti-Mouse IgG (H+L) Secondary Antibody, Qdot 705	Invitrogen	Cat#Q11062MP; RRID:AB_10393163
Monoclonal ANTI-FLAG M2 antibody produced in mouse	Sigma Aldrich	Cat#F3165; RRID:AB_259529
Anti-digoxigenin monoclonal antibody in rabbit	Thermo Fisher Scientific	Cat#9H27L19; RRID:AB_2532342
Monoclonal antibody S9.6	This Study	N/A
Bacterial and virus strains		
<i>E. coli</i> DH5 α	In-house made	N/A
<i>E. coli</i> lysogen	In-house made	N/A
<i>E. coli</i> BL21(DE3)	In-house made	N/A
<i>E. coli</i> DH10Bac	In-house made	N/A
Chemicals, peptides, and recombinant proteins		
FreeStyle 293 Expression Medium	Thermo Fisher Scientific	Cat#12338018
EX-CELL 405 Serum-Free Medium for Insect Cells	SAFC	Cat#14405C
Sf-900 III SFM	Thermo Fisher Scientific	Cat#12658027
cOmplete Protease Inhibitor Cocktail, EDTA-Free, Tablets	Sigma Aldrich	Cat#11873580001
Klenow Fragment (3-5' exo-)	New England Biolabs	Cat#M0212L
Albumin, biotin labeled bovine, lyophilized powder	Sigma Aldrich	Cat#A8549
Zinc sulfate monohydrate, $\geq 99.9\%$ trace metals basis	Sigma Aldrich	Cat#307491
18:1 Biotinyl Cap PE	Avanti Polar Lipids	Cat#870273P
18:1 PEG2000 PE	Avanti Polar Lipids	Cat#880130P
18:1 ($\Delta 9$ -Cis) PC (DOPC)	Avanti Polar Lipids	Cat#850375P
T3 RNA Polymerase	Promega	Cat#P2083
Biotin-16-dCTP	Jena Bioscience	Cat#NU-809-BIO16-S
DTT	Goldbio	Cat#DTT50
T4 DNA Ligase	New England Biolabs	Cat#M0202S
Lambda DNA	New England Biolabs	Cat#N3011S
Streptavidin	Thermo Fisher Scientific	Cat#434302
NEBNext Ultra II Q5 Master Mix	New England Biolabs	Cat#M0544S
Adenosine 5'-diphosphate sodium salt	Sigma Aldrich	Cat# A2754
BS3 (bis(sulfosuccinimidyl)suberate)	Thermo Fisher Scientific	Cat# 21580
Pierce TCEP-HCl	Thermo Fisher Scientific	Cat# 20491
SNAP-Surface Alexa Fluor 647	New England Biolabs	Cat# S9136S
Adenosine 5'-triphosphate disodium salt	Sigma Aldrich	Cat#A2383
Catalase	Sigma Aldrich	Cat#C100
Glucose Oxidase	Sigma Aldrich	Cat#G2133

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
RNase A	New England Biolabs	Cat#T3018L
RNase H	New England Biolabs	Cat#M0297S
Genomic-tip 500/G	Qiagen	Cat#10262
SYTOX Orange Nucleic Acid Stain	Thermo Fisher Scientific	Cat#S11368
Critical commercial assays		
HiScribe T7 High Yield RNA Synthesis Kit	New England Biolabs	Cat#E2040S
Wizard SV Gel and PCR Clean-Up System	Promega	Cat# A9281
HiFi DNA Assembly Master Mix	New England Biolabs	Cat#E2621S
QIAprep Spin Miniprep Kit	Qiagen	Cat#27106
Deposited data		
Original microscopy images used in figures	This study	Mendeley data: https://data.mendeley.com/datasets/2b72tw6wcv/1
Videos	This study	Mendeley data: https://data.mendeley.com/datasets/2b72tw6wcv/1
Cryo-EM map of cohesin-CTCF complex	This study	EMDB: EMD-32252
Coordinates of cohesin-CTCF complex	This study	PDB: 7W1M
Experimental models: Cell lines		
FreeStyle 293-F Cells	Thermo Fisher Scientific	Cat#R79007
Sf9 cells	Thermo Fisher Scientific	Cat# 12659017
High Five cells	Thermo Fisher Scientific	Cat# B85502
Oligonucleotides		
DNA oligos for molecular cloning	Integrated DNA Technologies	Table S2
Single guide RNAs	Synthego	Table S3
Recombinant DNA		
pBIG1a-SMC1A & SMC3	Kim et al. ³⁶	N/A
pBIG1a-SMC1A (E1157Q) & SMC3 (E1144Q)	Kim et al. ³⁶	N/A
pFastBac Dual-RAD21 (R172/D279/R450A)-TEV-MBP	Kim et al. ³⁶	N/A
pFastBac-STAG1-3C-TwinStrepII-His	Kim et al. ³⁶	N/A
pFastBac-STAG1 (W337A/F347A)-3C-TwinStrepII-His	This study	N/A
pFastBac-STAG1-TwinStrepII-SNAPf-His	Kim et al. ³⁶	N/A
pFastBac-His-TEV-MBP-3C-NIPBL (1163-2804)	Kim et al. ³⁶	N/A
pFastBac Dual-CTCF-TEV-MBP-FLAG	This study	N/A
pFastBac Dual-CTCF (266-580)-TEV-MBP-FLAG	This study	N/A
pFastBac Dual-CTCF (Y226A/F228A)-TEV-MBP-FLAG	This study	N/A
pFastBac Dual-CTCF-DM (251-264)-TEV-MBP-FLAG	This study	N/A
CMV-Intron-CTCF-TEV-MBP-FLAG	This study	N/A
6xHis-MBP-TEV-3xFLAG-dCas9	Jones et al. ⁷⁶	N/A
6xHis-Twin-Strep-SUMO-3xFLAG-dCas12a	Strohkendl et al. ⁷⁷	N/A
Software and algorithms		
Fiji	Fiji team	https://fiji.sc/
R	R core team	https://www.r-project.org

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Matlab	Matlab team	https://www.mathworks.com/products/matlab.html
Jupyter Notebook	Jupyter team	https://jupyter.org/ ;
Python	Python team	https://www.python.org/
Bowtie2	Langmead and Salzberg ⁷⁸	http://bowtie-bio.sourceforge.net/bowtie2/index.shtml
Samtools	Li et al. ⁷⁹	http://samtools.sourceforge.net/
Galaxy server	The Galaxy Community ⁸⁰	https://usegalaxy.org/
SerialEM	Mastronarde ⁸¹	https://bio3d.colorado.edu/SerialEM/
RELION-3	Zivanov et al. ⁸²	https://relion.readthedocs.io/en/release-3.1/
MotionCorr2	Zheng et al. ⁸³	https://emcore.ucsf.edu/ucsf-software
Gctf	Zhang ⁸⁴	https://www2.mrc-lmb.cam.ac.uk/download/gctf/
crYOLO	Wagner et al. ⁸⁵	https://cryolo.readthedocs.io/en/stable/
SWISS-MODEL server	Waterhouse et al. ⁸⁶	https://swissmodel.expasy.org
Phenix	Adams et al. ⁸⁷ ; Afonine et al. ⁸⁸	https://phenix-online.org
Coot	Emsley et al. ⁸⁹	https://www2.mrc-lmb.cam.ac.uk/personal/pemsley/coot/
UCSF Chimera	Pettersen et al. ⁹⁰	https://www.cgl.ucsf.edu/chimera/
UCSF ChimeraX	Goddard et al. ⁹¹	https://www.cgl.ucsf.edu/chimerax/
PyMOL	Schrodinger	https://pymol.org/2/
Hi-C data analysis pipeline	This study	https://github.com/mirnylab/distiller-nf ; version 0.0.3

RESOURCE AVAILABILITY**Lead contact**

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Ilya J. Finkelstein (ilya@finkelsteinlab.org).

Materials availability

Material generated in this study is available from the [Lead contact](#).

Data and code availability

- All data including Cryo-EM maps and coordinates have been deposited in the Electron Microscopy Data Bank EMDB: EMD-32252 and the Protein Data Bank PDB: 7W1M). Imaging data have been deposited at Mendeley Data and are publicly available as of the date of publication. All data are publicly available as of the date of publication. Accession numbers and DOI are listed in the [key resources table](#).
- This paper does not report any original code.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS**Bacterial strains**

dCas9 and dCas12a were expressed in *E. coli* BL21(DE3) using IPTG supplemented with the respective antibiotics at 18 °C. For recombineering λ-DNA, *E. coli* lysogen cells were grown at 30 °C.

Insect cell lines

Baculoviruses were made and amplified in Sf9 cells (Thermo Fisher Scientific) maintained in Sf-900™ III SFM (Thermo Fisher Scientific). Cohesin subunits and CTCF were expressed in High Five cells (Thermo Fisher Scientific) that were cultured in EX-CELL® 405 Serum-Free Medium for Insect Cells (SAFC).

FreeStyle 293-F cell culture

For making non-tagged CTCF, CTCF was expressed in FreeStyle 293-F Cells (Thermo Fisher Scientific). Cells were grown in suspension culture in FreeStyle 293 Expression Medium (Thermo Fisher Scientific).

METHOD DETAILS**Protein expression and purification****Cohesin-NIPBL complex**

The human cohesin-NIPBL complex was expressed and purified as described previously.^{36,43} Briefly, SMC1, SMC3, RAD21 (R172/D279/R450A) with a C-terminal MBP tag, and NIPBL^C (residues 1163-2804) with an N-terminal His-MBP tag were co-expressed in High Five cells (Thermo Fisher Scientific). SMC1, SMC3, RAD21 and NIPBL^C (cohesin-NIPBL^C without STAG1) were co-purified by Amylose Resin (NEB), a HiTrap Heparin HP column (GE Healthcare), a Resource Q column and a Superose 6 Increase 10/300 GL column (GE Healthcare). STAG1 with a C-terminal TwinStrepII-His dual tag or with a SNAPf-TwinStrepII-His tag was expressed separately. STAG1 was purified by Ni²⁺-NTA agarose (Qiagen) followed by a Resource Q column (GE Healthcare). The C-terminal dual tag in STAG1 was removed by home-made 3C protease. To form complete cohesin-NIPBL^C complex, STAG1 protein was added before Heparin column or Superose 6 Increase 10/300 GL column. Cohesin mutants, cohesin-WFA and cohesin-EQ were purified as wild-type proteins. Purified complexes were concentrated, flash frozen in liquid nitrogen and stored at -80 °C.

CTCF

Full-length CTCF was cloned into a modified pFastbac Dual vector (Thermo Fisher Scientific) with a C-terminal MBP and FLAG dual tag and expressed in High Five cells. Cells were resuspended in Lysis Buffer TN500 (20 mM Tris pH 7.5, 500 mM NaCl, 5% glycerol, 5 mM 2-mercaptoethanol) plus 25 μ M ZnSO₄, 1 $\times$ Pierce™ EDTA-free Protease Inhibitor Tablets (Thermo Fisher Scientific), and 1 mM phenylmethylsulfonyl fluoride (PMSF), and lysed by a high-pressure homogenizer (Microfluidics LM-20). After the addition of 0.3% polyethylenimine, cell lysate was cleared by centrifugation at 40,000 g for 1 h at 4 °C. The supernatant was mixed with Amylose Resin for 1.5 h in the cold room. The resin was washed with Lysis Buffer TN500 plus 500 mM NaCl and 25 μ M ZnSO₄, followed by the Buffer TN150 (20 mM Tris pH 8.0, 150 mM NaCl, 5% glycerol, 5 mM 2-mercaptoethanol) plus 25 μ M ZnSO₄, and eluted by the Buffer TN150 plus 40 mM maltose and 25 μ M ZnSO₄. The protein was further purified by a Resource Q column and finally a Superose 6 Increase 10/300 GL column.

To reconstitute the CTCF-DNA complex, purified CTCF and dsDNA (GCAAGATTGCAGTGCCACAGAGGCCAGCAGG GGGCGCTAGTGAGGTGGTTTTATATGTTTTGTTATGTATTGTTATTTCCCTTAATTTTAGGATATGAAAAACAAGAATTTATC; underlined sequence is CTCF-binding site, CBS) were incubated at equal molar ratio on ice for 1 h, and then loaded onto a Superose 6 Increase 10/300 GL. Purified samples were concentrated, flash-frozen in liquid nitrogen and stored at -80 °C. CTCF mutants, CTCF-YFA, CTCF-ZF (266-580) were purified as wildtype proteins.

The untagged CTCF was expressed in and purified from FreeStyle™ 293-F cells. Full-length CTCF was cloned into a mammalian vector with a C-terminal MBP and FLAG dual tag and expressed in 500 mL FreeStyle™ 293-F cells. Cells were collected 48 hours after transfection. To purify CTCF, cells were resuspended in Lysis Buffer TN500 (20 mM Tris pH 7.5, 500 mM NaCl, 5% glycerol, 5 mM 2-mercaptoethanol) plus 25 μ M ZnSO₄, 1 $\times$ Pierce™ EDTA-free Protease Inhibitor Tablets, and 1 mM PMSF, and lysed by mild sonication. After the addition of 0.3% polyethylenimine, cell lysate was cleared by centrifugation at 40,000 g for 1 h at 4 °C. The supernatant was mixed with Amylose Resin for 1.5 h in the cold room. The resin was washed with Lysis Buffer TN500 plus 500 mM NaCl and 25 μ M ZnSO₄, followed by the Buffer TN150 (20 mM Tris pH 8.0, 150 mM NaCl, 5% glycerol, 5 mM 2-mercaptoethanol) plus 25 μ M ZnSO₄, and eluted by the Buffer TN150 plus 40 mM maltose and 25 μ M ZnSO₄. 500 μ g of TEV protease was added into protein and incubated overnight at 4 °C. After that, the protein was further purified by a His Trap Heparin column and finally a Superdex200 10/300 GL column (GE Healthcare), and stored in the storage buffer (20 mM HEPES pH 7.6, 150 mM KCl, 1 mM DTT, 0.01% IGEPAL CA-630, 50 μ M ZnSO₄).

The MBP-FLAG tagged CTCF-DM(251-264) was expressed and purified from High Five cells as described for the human construct.

Nuclease-dead (dCas9)

The fusion construct of nuclease-dead *S. pyogenes* Cas9 (dCas9) with D10A/H840A mutations contained an N-terminal hexahistidine-maltose binding protein (His₆-MBP) tag and a peptide sequence containing a tobacco etch virus (TEV) protease cleavage site,⁹² followed by triple FLAG epitope tag. Protein was expressed in BL21 star (DE3) cells and purified as described previously.⁷⁶ The detailed processes of expression and purification are as follows. A 25 mL culture of LB medium with 50 mg mL⁻¹ kanamycin was inoculated with a single colony and grown overnight at 30 °C with shaking. 1 L of LB with 50 mg mL⁻¹ kanamycin was inoculated with 10 mL of the starter culture and then grown to an OD₆₀₀ of 0.6 at 30 °C. Protein expression was induced with 1 mM IPTG for 18 h at 18 °C. Cells were collected by centrifugation and lysed by sonication with 45% amplitude of ultrasound at 4 °C in lysis buffer (20 mM Na-HEPES, pH 8.0, 1 M NaCl, 1 mM EDTA, 5% glycerol, 0.1% Tween-20, 1 mM PMSF, 200 U DNase I (NEB), 1 $\times$ HALT protease inhibitor (Thermo Fisher Scientific). The lysate was clarified by ultracentrifugation at 45,000 relative centrifugal force (RCF) and then passed over a nickel affinity column (HisTrap FF 5 mL, GE Healthcare) and eluted with elution buffer (20 mM Tris-Cl, pH 8.0, 250 mM NaCl, 250 mM imidazole). The His₆-MBP was cleaved overnight in dialysis buffer (20 mM HEPES-KOH, pH 7.5, 150 mM KCl, 10% glycerol, 1 mM DTT, 1 mM EDTA) in the presence of homemade TEV protease (0.5 mg per 50 mg of protein). The dialyzed

protein was resolved on a HiTrap SP FF 5 mL column (GE Healthcare) with a linear gradient between buffer A (20 mM HEPES-KOH, pH 7.5, 100 mM KCl) and buffer B (20 mM HEPES-KOH, pH 7.5, 1 M KCl). Protein-containing fractions were concentrated via dialysis (10 kDa Slide-A-Lyzer, Thermo Fisher Scientific) and then passed over a Superdex 200 Increase 10/300 column (GE Healthcare) pre-equilibrated into storage buffer (20 mM HEPES-KOH, pH 7.5, 500 mM KCl). The protein was snap frozen in liquid nitrogen and stored in 10 μ L aliquots at -80°C .

Nuclease-dead Cas12a (Cas12a)

The nuclease-dead *Acidaminococcus* sp. Cas12a (dCas12a) containing D908A point mutant was expressed and purified using a previously established protocol with minor modifications.⁷⁷ The construct of dCas12a contained a His₆-Twin-Strep-SUMO N-terminal fusion and a triple FLAG epitope C-terminal fusion was transformed and expressed in BL21 star (DE3) cells. A 20 mL culture of Terrific Broth (TB) with 50 mg mL⁻¹ carbenicillin was inoculated with a single colony and grown overnight at 37 $^{\circ}\text{C}$ with shaking. 1 L of TB with 50 mg mL⁻¹ carbenicillin was inoculated with 10 mL of the starter culture and then grown to an OD₆₀₀ of 0.6 at 37 $^{\circ}\text{C}$. Protein expression was induced with 0.5 mM IPTG for 24 h at 18 $^{\circ}\text{C}$. Cells were collected by centrifugation at 45,000 RCF and lysed by sonication with 45% amplitude of ultrasound at 4 $^{\circ}\text{C}$ in lysis buffer (20 mM Na-HEPES, pH 8.0, 1 M NaCl, 1 mM EDTA, 5% glycerol, 0.1% Tween-20, 1 mM PMSF, 200 U DNase I, 1 $\times$ HALT protease inhibitor). The supernatant clarified by ultracentrifugation at 45,000 RCF was loaded to a hand-packed StrepTactin Superflow gravity column (IBA Lifesciences) and then eluted (20 mM Na-HEPES, 1 M NaCl, 5 mM desthiobiotin, 5 mM MgCl₂, 5% glycerol). The eluate was concentrated using a 30-kDa MWCO spin concentrator (Millipore), and 3 μ M SUMO protease was added and then incubated overnight on a rotator at 4 $^{\circ}\text{C}$. The cleaved protein was then fractionated over a HiLoad 16/600 Superdex 200 Column (GE Healthcare) pre-equilibrated with storage buffer (20 mM Na-HEPES, 150 mM KCl, 5 mM MgCl₂, 2 mM DTT buffer). The protein was finally snap frozen in liquid nitrogen and stored in 10 μ L aliquots at -80°C .

Recombineering lambda DNA containing CTCF binding sites

The DNA substrate is derived from bacteriophage λ . We recombineered λ phage lysogens in *E. coli* using the Red system as described previously with the following modification.³⁴ DNA cassettes with up to four CTCF binding sites were ordered from Integrated DNA Technologies (IDT). A 5 mL LB culture of lysogen cells (IF189) transformed with pKD78 plasmid (pIF284) containing Red recombinase system were grown overnight at 30 $^{\circ}\text{C}$ in the presence of 10 μ g μ L⁻¹ chloramphenicol. 350 μ L of cells were used to inoculate a fresh 35 mL culture of LB containing the same concentration of antibiotic. When the cells reached an OD₆₀₀ $\sim$ 0.5, the Red recombinase system was induced by adding 2% L-arabinose (GoldBio) and incubated for an additional 1 hour at 30 $^{\circ}\text{C}$. Cells were harvested at 4,500 RCF for 7 min, washed three times in ice-cold Milli-Q H₂O, and finally resuspended in 200 μ L of H₂O. The cells were kept on ice and used immediately for the recombineering reaction.

For recombineering, the resuspended lysogen cells were mixed with 100 ng of the insertion cassette (Table S8) and were electroporated at 12.5 kV cm⁻¹ in 0.2 cm cuvette using a MicroPulser Electroporator (Bio-Rad 165-210). Cells were immediately resuspended in 1 mL of SOC and then transferred to a culture tube containing 2 mL LB broth. After at least 6 hours outgrowth at 30 $^{\circ}\text{C}$, 300 μ L of the culture was plated onto LB agar plates containing 30 μ g mL⁻¹ carbenicillin. Successful incorporation of recombinant DNA was checked via colony PCR with oligos IF751 (GAA CAA ACA ATA CCC AGA TTG CG) and oligo IF752 (GGA ATA TCT GGC GGT GCA AT) and were further confirmed by DNA sequencing.

To purify these DNA substrates from recombinant phages, we induced phage production via heat shock. A single lysogenic colony was grown in 50 mL of LB broth with 50 μ g mL⁻¹ carbenicillin overnight at 30 $^{\circ}\text{C}$. 5 mL of this starter culture was used to inoculate 500 mL of LB on the following day. When the cells reached an OD₆₀₀ $\sim$ 0.6, they were incubated in a 70 $^{\circ}\text{C}$ water bath to rapidly increase the temperature to 42 $^{\circ}\text{C}$. The culture was placed at 45 $^{\circ}\text{C}$ in a shaking incubator for 15 minutes and then transferred to a 37 $^{\circ}\text{C}$ incubator for 2 hours. To liberate the phage particles, cells were harvested by centrifugation at 3,000 RCF for 30 minutes and lysed via resuspension in 10 mL of SM buffer (50 mM Tris-HCl pH 7.5, 100 mM NaCl, 8 mM MgSO₄) with 2% chloroform, and rotated at 37 $^{\circ}\text{C}$ for 30 min. A subsequent 1 hour incubation with 50 ng μ L⁻¹ DNase I (Sigma-Aldrich D2821) and 30 ng μ L⁻¹ RNaseA (Sigma-Aldrich R6513) degraded the bacterial genomic DNA and RNA. The clarified lysate, containing soluble phage capsids, was obtained by centrifugation for 15 minutes at 6,000 RCF at 4 $^{\circ}\text{C}$, and further diluted with 40 mL of SM buffer. Phage capsids were precipitated by incubating for 1 hour with 10 mL ice-cold buffer L2 (30% PEG 6000, 3 M NaCl) and then harvested by centrifugation at 10,000 RCF for 10 minutes at 4 $^{\circ}\text{C}$. The phage pellet was washed with 1 mL of buffer L3 (100 mM Tris-HCl pH 7.5, 100 mM NaCl, 25 mM EDTA) and then resuspended with 3 mL of buffer L3, followed by an equal volume of buffer L4 (4% SDS). The phage capsid proteins were further digested by incubation with 100 ng μ L⁻¹ of proteinase K (NEB #P8012S) for 1 hour at 55 $^{\circ}\text{C}$. SDS was precipitated with 3 mL buffer L5 (3 M potassium acetate pH 7.5), and the cloudy solution was clarified by centrifugation at 15,000 RCF for 30 minutes at 4 $^{\circ}\text{C}$. The soluble phage DNA was passed over a pre-equilibrated Qiagen tip-500 column (Qiagen #10262), washed with buffer QC (1.0 M NaCl, 50 mM MOPS pH 7.0, 15% isopropanol) and eluted with 15 mL buffer QF (1.25 M NaCl, 50 mM Tris pH 8.5, 15% isopropanol). Finally, DNA was precipitated with the addition of 10.5 mL of 100% isopropanol, rinsed in 70% ethanol twice and dissolved in 500 μ L TE buffer (10 mM Tris-HCl pH 8.0, 1 mM EDTA). DNA was stored at 4 $^{\circ}\text{C}$.

Single-molecule fluorescence microscopy

Flowcell preparation

Flowcells used for single-molecule DNA experiments were prepared as previously described.⁹³ Briefly, a 4-mm-wide, 100- μ m-high flow channel was constructed between a glass coverslip (VWR 48393 059) and a custom-made quartz microscope slide patterned

with 1–2 μm Chromium barriers using two-sided tape (3M 665). The surface was passivated with a fluid biotinylated lipid bilayer. All experiments were conducted at 37 °C under indicated flow rate. Single-molecule fluorescent images were collected with a customized prism TIRF microscopy-based inverted Nikon Ti-E microscope system. The sample was illuminated with a 488 nm laser (Coherent Sapphire) and a 637 nm laser (Coherent OBIS) through a quartz prism (Tower Optical Co.). For imaging SYTOX Orange-stained DNA and Alexa488-labeled CTCF, the 488 nm laser power was adjusted to deliver low power (4 mW) at the front face of the prism using a neutral density filter set (Thorlabs). For imaging Alexa647-labeled cohesin, the 637 nm laser power was adjusted to 10 mW. Multi-color imaging was recorded using electron-multiplying charge-coupled device (EMCCD) cameras (Andor iXon DU897). DNA was visualized by a continuous flow (0.12 mL min⁻¹) in the imaging buffer (40 mM Tris-HCl pH 8.0, 2 mM MgCl₂, 0.2 mg mL⁻¹ BSA, 50 mM NaCl, 1 mM DTT, and 1 mM ATP, 100 nM SYTOX Orange) supplemented with an oxygen scavenging system (3% D-glucose (w/v), 1 mM Trolox, 1500 units catalase, 250 units glucose oxidase; all from Sigma-Aldrich). Unless indicated, 1 mM ATP was added in the imaging buffer. NIS-Elements software (Nikon) was used to collect the images at 2–3 s frame rate with 150 ms exposure time. All images were exported as uncompressed TIFF stacks for further analysis in FIJI (NIH) and MATLAB (The MathWorks).

Preparation of single-tethered CTCF-DNA substrates

To prepare DNA substrates for microscopy, 125 μg of λ -phage CTCF-DNA was mixed with two oligos (2 μM oligo Lab07 and 2 μM oligo Lab09 for CTCF^N-DNA, or 2 μM / oligo Lab06 and 2 μM oligo Lab08 for CTCF^C-DNA) in 1 $\times$ T4 DNA ligase reaction buffer (NEB B0202S) and heated to 70 °C for 15 min followed by gradual cooling to 15 °C for 2 hours. One oligo will be annealed with the overhang located at the left cohesive end (*cosL*) of DNA, and the other oligo will be annealed with the overhang at right cohesive end (*cosR*). After the oligomer hybridization, 2 μL of T4 DNA ligase (NEB M0202S) was added to the mixture and incubated overnight at room temperature to seal nicks on DNA. The ligase was inactivated with 2 M NaCl, and the reaction was injected to an S-1000 gel filtration column (GE) to remove excess oligonucleotides and proteins.

Preparation of U-shaped CTCF-DNA substrates

The strategy of making U-shaped double-tethered DNA was described previously.³⁶ Briefly, Two biotinylated oligos (2 μM oligo Lab06 and 2 μM oligo Lab07) were annealed and ligated to CTCF-DNA as mentioned above. The biotin-BSA stock solution (10 mg mL⁻¹, Sigma-Aldrich A8549) was prepared in the T50 buffer (10 mM Tris, 50 mM NaCl, pH 8.0) and stored at 4 °C. The solution was diluted 10 times with the T50 buffer right before use and was injected into the flowcell. After 10 minutes BSA coating, 300 μL streptavidin solution (0.1 mg mL⁻¹, Invitrogen 434301) was loaded into the flowcell and incubated for 10 minutes at room temperature. For further passivation of the surface to prevent nonspecific adsorption of nucleic acids and proteins, 30 μL of the liposome stock solution (Avanti Polar Lipids; 98 mol% DOPC, 2 mol% DOPE-mPEG2K) was diluted in 800 μL of the lipid buffer (10 mM Tris pH 8.0, 100 mM NaCl) and loaded into the flowcell. After 20 minutes incubation, 20 μL of CTCF-DNA stock was diluted in 1 mL T50 buffer and incubated for 5 min in the chamber. The flowcell was then immediately used for the single-molecule imaging on the TIRF microscope.

Preparation of R-loops DNA substrates

The template DNA used for R-loop DNA substrates, pFC53-Airn plasmid (3991 bp), containing the mouse Airn sequence with downstream of a T3 promoter, was a generous gift from the Chedin lab.⁶¹ The process of making long R-loop DNA was based on the previous protocol with minor modifications.⁵⁹ Briefly, 600 μg template DNA plasmid was linearized with 4 μL restriction enzyme BsaI (NEB), which was then inactivated by heat. The linear DNA was ligated using Quick Ligation Kit (NEB) for overnight incubation at room temperature to form long ligated DNA. Biotinylated nucleotide was synthesized to one end of long ligated DNA. For the reaction, 25 μM dATP (NEB), 25 μM dTTP (NEB), 25 μM dGTP (NEB), 5 μM dCTP (NEB), 20 μM bio-dCTP (Jena Bioscience, NU-809-BIO-16-S) and 2 units Klenow Fragment 3' $\rightarrow$ 5' exo- (NEB M0212S) were added into the ligated DNA product in the presence of 1X T4 DNA ligase buffer (NEB B0202S). After 30 minutes incubation at 25 °C, the DNA was purified using phenol/chloroform extraction and ethanol precipitation. The purified biotinylated long ligated DNA was stored at -20 °C. The long R-loops DNA was generated by *in vitro* transcription, which was carried out at 37 °C for 1 hour in 1X Transcription Optimized Buffer (Promega, P4024) with 120 μg biotinylated long ligated DNA, 4 μL T3 RNA polymerase (Promega, P4024), 20 mM DTT, 0.05% Tween-20, and 500 μM rNTP. Transcription was terminated by heat inactivation at 65 °C for 10 minutes. Before R-loops DNA was injected into flowcell, 1 μL RNase A (NEB T3018L) was added to degrade free RNA at 37 °C for 30 minutes incubation. For control DNA without R-loops, R-loops DNA was treated by adding 2 μL RNase H (NEB M0297S) at 37 °C for 30 minutes incubation. This protocol produces short biotinylated fragments that decohere the flowcell surface, bind cohesin, and reduce the signal to background for longer R-loop concatamers. To reduce the background signal from short biotinylated DNA fragments, 1 μm diameter streptavidin-coated beads with (NEB S1420S) were injected into the flowcell prior to R-loop DNA injection. Longer DNA molecules that bind these beads are easily distinguishable from short DNA fragments. By keeping these long DNAs on the surface, we reduce the background signal from surface-bound short DNAs. Beads that had two or more DNA molecules were not included in our analysis. We only focus on the single R-loop DNA molecule. Monoclonal antibody S9.6 (gift from Paull Lab) coupled with Goat anti-mouse Alexa Fluor 488 (Thermo Fisher, A28175) was used to detect R-loop.

Imaging cohesin and CTCF on DNA

For imaging Alexa647-labeled wild type cohesin, 1 μM of the protein complex was diluted to 7 nM in a total volume of 150 μL imaging buffer. For the fluorescent labeling of cohesin mutants, cohesin-WFA and cohesin-EQ, 7 nM of protein were pre-incubated with 10 nM anti-His6 antibody (Takara, 631212) and 12 nM anti-mouse quantum dots (QD₇₀₅) (Thermo Fisher, Q11062MP) on ice for 10 minutes. The mixture was then diluted to a total volume of 150 μL imaging buffer. For the fluorescent labeling of CTCF,

CTCF-ZF, CTCF-YFA, 2 nM of protein were conjugated with 4 nM monoclonal anti-FLAG antibody (Sigma Aldrich, F3165) and 6 nM Goat anti-mouse Alexa Fluor 488 on ice for 10 minutes. For U-shaped DNA experiments shown in Figure S3B, 2 nM CTCF were conjugated with 4 nM monoclonal anti-FLAG antibody and 6 nM Goat anti-Mouse QDot 705 on ice for 10 minutes before flowcell injection. The mixture was then diluted to a total volume of 150 μ L imaging buffer. Immediately after the conjugation or dilution, the fluorescently labeled proteins were injected into the flowcell at a 0.12 mL min⁻¹ flow rate. 100 μ L of 5 ng/mL Heparin was loaded into the flowcell prior to CTCF to remove non-specific CTCF binding.

dCas9-RNP and dCas12a-RNP preparation and labeling

dCas9 and dCas12 ribonucleoprotein (RNP) complexes were reconstituted by incubating a 1:2 molar ratio of apoprotein and RNA (see Table S3 for sgRNA sequences) in incubation buffer (50 mM Tris-HCl pH 8.0, 5 mM MgCl₂, 2 mM DTT) followed by incubation at 25 °C for 25 minutes. dCas9 and dCas12a were labeled by anti-FLAG coupled with Alexa Fluor 488, and then diluted to 5 nM–10 nM in imaging buffer before injected into the flowcell.

Cryo-electron microscopy

Sample preparation and data acquisition

For cryo-EM sample preparation, purified cohesin-NIPBL^C complex and CTCF-dsDNA complex were incubated in 20 mM HEPES pH 7.5, 50 mM KCl, 5% glycerol, 1 mM DTT, 2 mM MgCl₂, 0.01% IGEPAL® CA-630 for 30 min on ice. Then 1 mM ADP, 1 mM BeSO₄ and 8 mM NaF were added, and the sample was incubated at 30 °C for 25 min. For crosslinked sample, a further 1 mM bis(sulfosuccinimidyl)suberate (BS³) was added to the sample and incubated for 30 min on ice. Both samples were applied to a 20–40% sucrose gradient in 20 mM HEPES pH 7.5, 100 mM KCl, 0.5 mM TCEP, 2 mM MgCl₂, 0.01% IGEPAL® CA-630, and ultracentrifugation was carried out at 36,000 rpm for 16 h at 4 °C. Samples were analyzed by 6% SDS-PAGE. The fractions containing cohesin-NIPBL-CTCF-DNA complex were pooled and concentrated. Then the sample buffer was exchanged to 20 mM HEPES pH 7.5, 100 mM KCl, 0.5 mM TCEP, 2 mM MgCl₂, 2 mM fluorinated Fos-Choline-8.

The freshly purified, crosslinked cohesin-NIPBL^C-CTCF-DNA complex (OD₂₈₀ = 7.5–10, OD₂₆₀/OD₂₈₀ = 1.25) was applied to glow-discharged Quantifoil R1.2/1.3 300-mesh gold holey carbon grids (Ted Pella). Grids were blotted for 2.5 s under 100% humidity at 4 °C before being plunged into liquid ethane using a Vitrobot Mark IV (FEI). Micrographs were acquired on a Titan Krios microscope (FEI) operated at 300 kV with a K3 Summit direct electron detector (Gatan), using a slit width of 20 eV on a GIF-Quantum energy filter. SerialEM software⁸¹ was used for automated data collection following standard procedures. A calibrated magnification of 46,296 X was used for imaging, yielding a pixel size of 1.08 Å on images. Before image processing, the images were binned further to reduce the particles image size, resulting the final pixel size of 1.44 Å. The defocus range was set from -1.5 μ m to -2.5 μ m. Each micrograph was dose-fractionated to 36 frames, with a total dose of about 60 e⁻/Å².

Cryo-EM image processing

The detailed image processing statistics are summarized in Figures S5 and S6 and Table S1. Motion correction was performed using the MotionCorr2 program,⁸³ and the CTF parameters of the micrographs were estimated using the Gctf program.⁸⁴ Most of steps of image processing were performed using RELION-3.^{82,94} Totally 9,121 movie frames were collected. Initially, particles picking was performed by crYOLO.⁸⁵ Class averages representing projections in different orientations selected from the initial 2D classification were used as templates for the reference based automatic particle picking. Extracted particles were binned 2 times and subjected to 2D classification. Particles from the classes with fine structural features were selected for 3D classification and reported cohesin-NIPBL-DNA structure was used as an initial model. Particles from one 3D classes showing good structural features were selected for second round classification with fine angular sampling and local search. The particles from the three good classes were chosen and subject to 3D refinement, CTF refinement and particle polishing, generating three maps with overall resolution of 3.7 Å, 7.0 Å and 6.5 Å, respectively.

To improve the resolution for CTCF and bound DNA, we performed focused 3D refinement with different sizes of soft masks. All resolutions were estimated by applying a soft mask around the protein density using the gold-standard Fourier shell correlation (FSC) = 0.143 criterion.⁹⁵ The local resolution was calculated with RELION-3.

Cryo-EM model building and refinement

Models of STAG1-RAD21-CTCF and CTCF ZF1 for docking into cryo-EM maps were generated by SWISS-MODEL server⁸⁶ using the structures of STAG2-RAD21-CTCF and CTCF ZFs-DNA complexes (PDB 6QNX, 5YEL) as templates.^{19,44,45} These models and reported structures of cohesin-NIPBL^C-DNA (PDB 6WGE) and CTCF ZFs-DNA (PDB 5T0U, 5YEL) complexes^{43–45} were docked onto cryo-EM maps in UCSF Chimera⁹⁰ and refined by real-space refinement in Phenix with rigid body and secondary structure restraints.^{87,88} Then model was iteratively manually built in coot⁸⁹ and refined in Phenix. Model validation was performed in Molprobit.^{96,97} Structural images were generated in UCSF Chimera,⁹⁰ ChimeraX⁹¹ and PyMOL (Schrodinger).

QUANTIFICATION AND STATISTICAL ANALYSIS

Fluorescent image analysis

The DNA compaction percentage and rate were analyzed in FIJI. The Mann-Whitney t-test was used to determine the significant difference between experimental conditions. Error bars on the binding distribution histogram were calculated in MATLAB using bootstrap analysis with replacement. The significance threshold was set at 0.05 in all tests.

The number of CTCF on CBS on CTCF-DNA was calculated by dividing the initial fluorescence intensity of each CTCF patch on CBS, which was averaged from first 5 frames within a 6 x 6 pixel region of interest (ROI), by the average initial intensity of CTCF patches on non-specific sites. To determine CTCF lifetime on DNA, we measured the total amount of time that each CTCF punctum spent on its DNA-binding site. Survival data were fit to single-exponential decays via a custom Matlab script. The boxplots plotted by Matlab denote the first and third quartiles of the data. The whiskers each cover 25% of the data values.

ChIP-seq and DRIP-seq analysis

Cohesin subunits ChIP-Seq dataset and R-loop DRIP-Seq dataset for mouse embryonic fibroblasts and human cell lines were mined from public GEO database^{62,63,66–68} (Tables S4 and S6). The raw ChIP-seq sequence data were downloaded using Faster Download and Extract Reads in BAM format from NCBI SRA on Galaxy server,^{80,98} and then were mapped against the human hg19 reference using bowtie2.⁷⁸ For the biological replicate experiments, the resulting alignments from each experiments were combined using samtools merge.⁹⁹ Peaks were called by MACS2 with the parameters settings described previously.^{62,100} The BED file of called R-loop peaks of mouse embryonic fibroblasts was directly downloaded from GEO database (GSM1720621). The peak overlaps were determined using BEDTools suite with default parameters.⁶⁴ The genomic annotation of peak overlaps was analyzed using ChIPseeker R package.¹⁰¹ The read density profiles and heatmaps of R-loop across overlapped regions were analyzed using deeptools.¹⁰² The significance of peak overlaps were calculated using bedtools fisher, ChIPseeker with nShuffle of 10000 and Genomic HyperBrowser with Monte Carlo sampling of 10000¹⁰³ (Tables S5 and S7).

Hi-C analysis

Hi-C data from MEFs were mapped at 1 kb resolution with the mm9 genome assembly and distiller pipeline (<https://github.com/mirnylab/distiller-nf>; version 0.0.3). The mapped data were converted to cooler files¹⁰⁴ and balanced by iterative correction.¹⁰⁵ Pile ups on R-loops were performed similarly to pile ups described previously.²⁰ Snippets of “observed” 5-kb-resolution Hi-C contact maps centered on R-loops, which were previously determined by DRIPc-seq.⁶³ Observed snippets were normalized by the contact probability scaling (“expected”) to produce “observed-over-expected” snippets. Observed-over-expected snippets were averaged together to produce the average Hi-C contact enrichment maps. Gene annotations from GENCODE (<https://www.genencodegenes.org/>) were used to identify intra- and intergenic regions.¹⁰⁶